FACULTY OF MECHANICAL ENGINEERING

CAN THO UNIVERSITY OF TECHNOLOGY

PHÁT TRIỂN ỨNG DỤNG IOT

(IoT application development)

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3.2: Transmission Control Protocol/Internet Protocol (TCP/IP)

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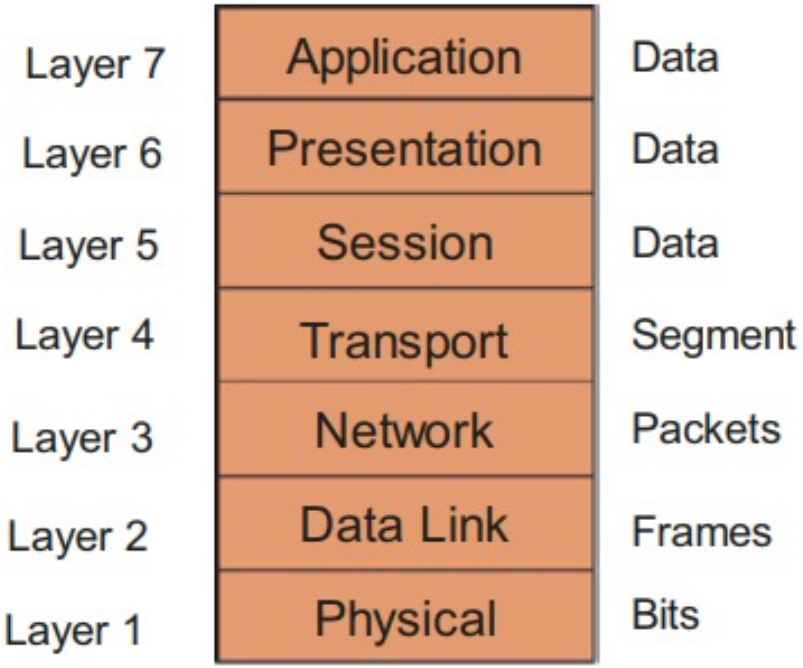
3.11: IoT Actuators

3.1.The Open System Interconnection Model

- Nearly all data networks in use today are based on the Open Systems Interconnection (OSI) standard.
- The OSI model was introduced by the International Organization for Standardization (ISO), in 1984, to address this complex problem. ISO is a global federation of national standards organizations representing over 100 countries.
- The model is intended to describe and standardize the main communication functions of any telecommunication or computing system without regard to their underlying internal structure and technology. Its goal is the interoperability of diverse communication systems with standard protocols.

3.1.The Open System Interconnection Model

- The OSI is a conceptual model of how various components communicate in data-based networks. It uses "divide and conquer" concept to virtually break down network communication responsibilities into smaller functions, called layers, so they are easier to learn and develop.



3.1.The Open System Interconnection Model

Summary of key functions, devices, and protocols of the OSI layers

OSI layer	Main function	Examples of main devices	Examples of main protocol
Application	Provides network services to the end-host's applications	Server, laptops, PCs	HTTPS, FTP, Telnet, SSH
Presentation	Ensures the data can be understood between two end hosts	N/A	Data encoding, data formatting, and serialization
Session	Manages multiple sessions between end hosts	N/A	Connection management, error recovery
Transport	Establishes end-to-end connectivity and ensures reliable data delivery	Firewalls	TCP, UDP
Network	Connectivity and path selection based on logical addresses	Routers, firewalls	IPv4, IPv6
Data link	Defines data format for transmission	Switches, APs	IEEE 802.1 (Ethernet), PPP
Physical	Defines physical media access and properties	Fiber optics, category 5 cables, coaxial cables	IEEE 802.3

Mô hình OSI là gì? Chức năng của các tầng giao thức trong mô hình OSI - TOTOLINK Việt Nam

3.1.The Open System Interconnection Model

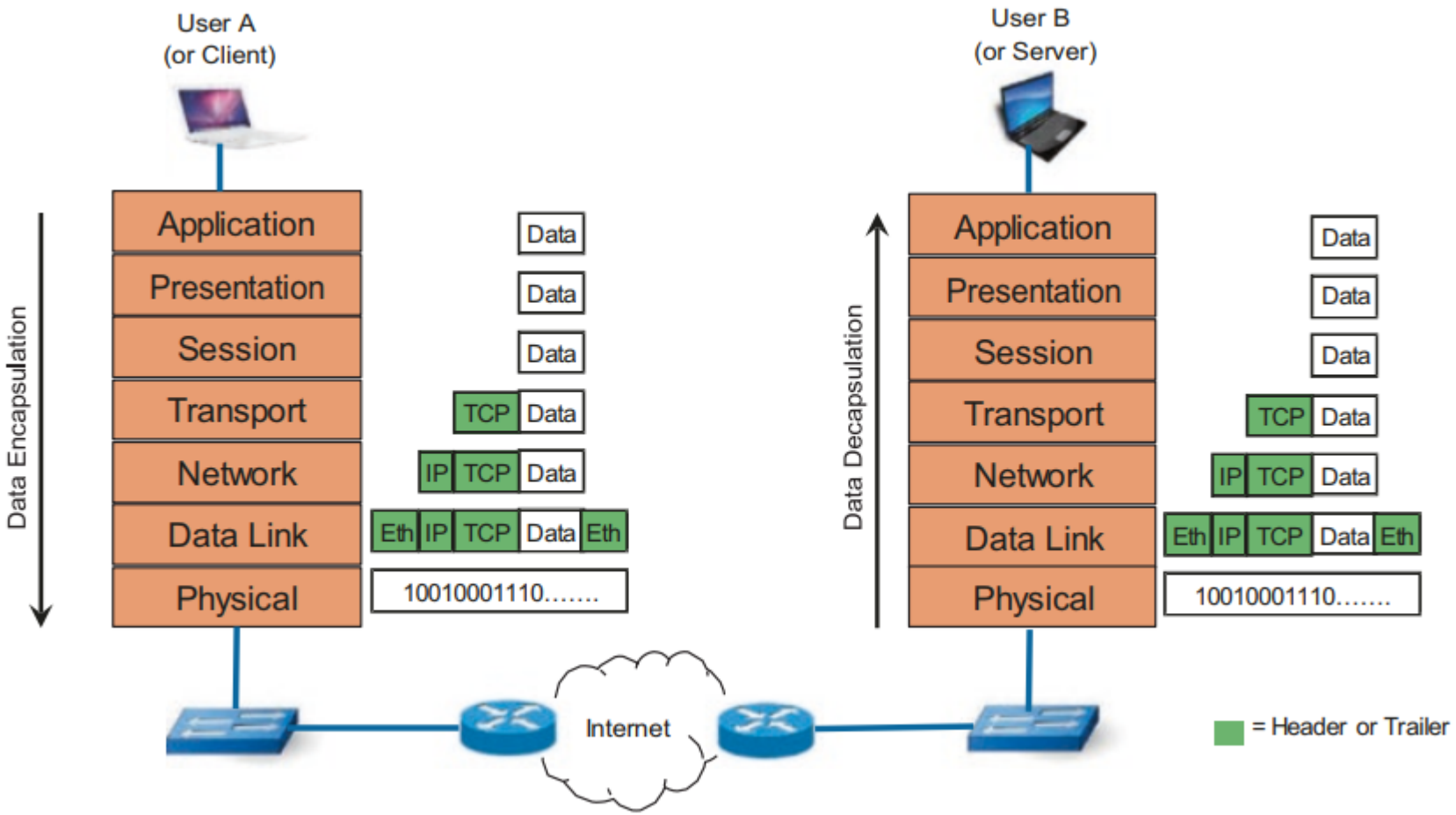
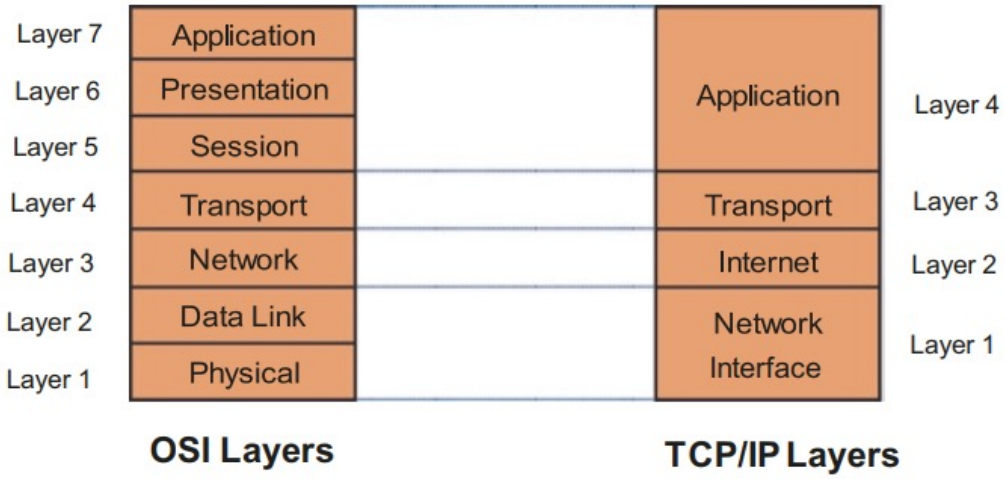


Illustration of OSI model

3.2.Transmission Control Protocol/Internet Protocol (TCP/IP)

- TCP/IP (Transmission Control Protocol/Internet Protocol) is a connection-oriented transport protocol suite that sends data as an unstructured stream of bytes.
- TCP/IP was in the process of development when the OSI standard was published in 1984. The TCP/IP model is not exactly the same as OSI model. OSI is a seven layered standard, but TCP/IP is a four-layered standard.

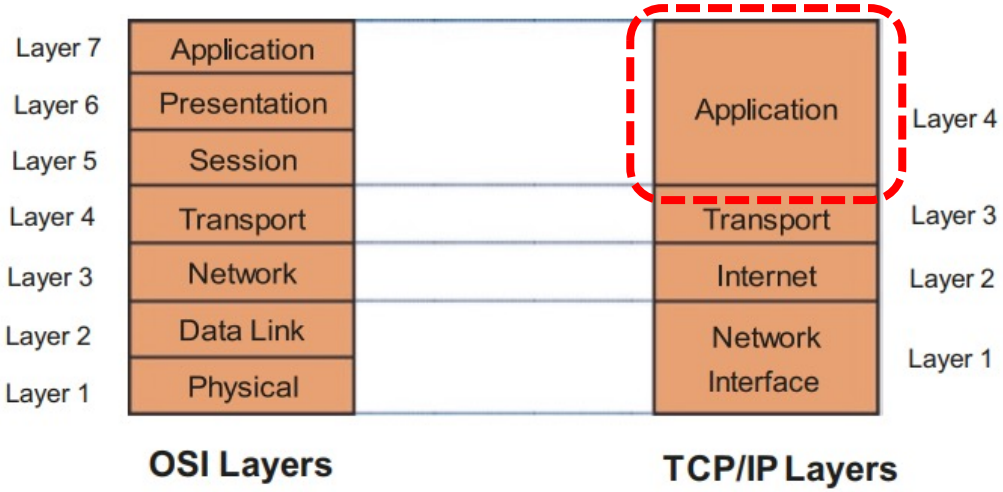


Relationship between OSI reference model and TCP/IP

3.2.Transmission Control Protocol/Internet Protocol (TCP/IP)

TCP/IP Layer 4: Application Layer

- As with the OSI model, the Application Layer is the topmost layer of TCP/IP model. It combines the Application, Presentation, and Session Layers of the OSI model.
- The Application Layer defines TCP/IP application protocols and how host programs interface with Transport Layer services to use the network.

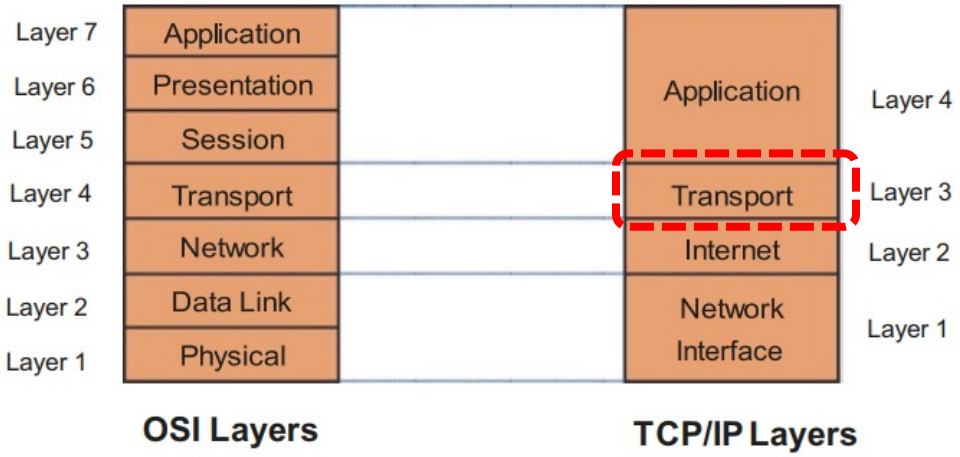


Relationship between OSI reference model and TCP/IP

3.2.Transmission Control Protocol/Internet Protocol (TCP/IP)

TCP/IP Layer 3: Transport Layer

- The Transport Layer is the third layer of the four-layer TCP/IP model. Its main tenacity is to permit devices on the source and destination hosts to carry on a conversation. The Transport Layer defines the level of service and status of the connection used when transporting data. The main protocols included at the Transport Layer are TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).

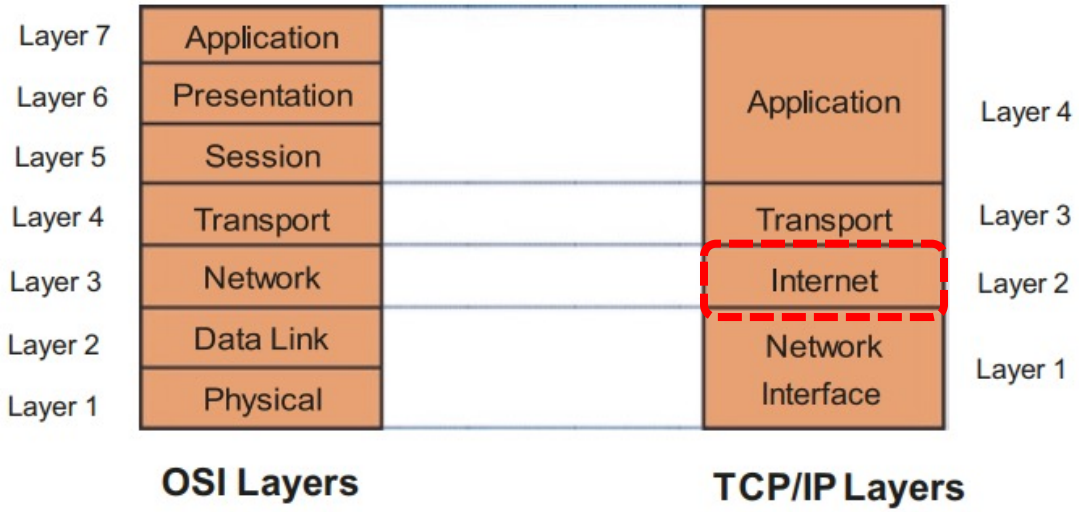


Relationship between OSI reference model and TCP/IP

3.2.Transmission Control Protocol/Internet Protocol (TCP/IP)

TCP/IP Layer 2: Internet Layer

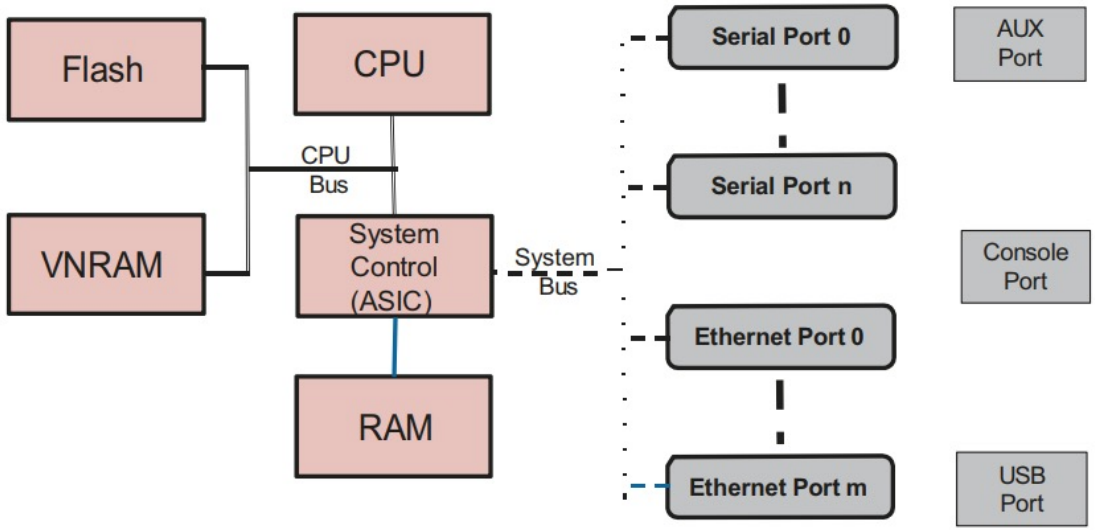
- The Internet Layer of the TCP/IP stack packs data into data packets known as IP datagrams, which contain source and destination address information that is used to forward the datagrams between hosts and across networks.



Relationship between OSI reference model and TCP/IP

Router Main Components

- *NVRAM (Nonvolatile RAM)*: NVRAM is used to store the startup configuration file "startup config," which is used during system startup to configure the software. This is due to the fact that NVRAM does not lose its content when the power is turned off. In other words, the router's configuration is not erased when the router is reloaded.

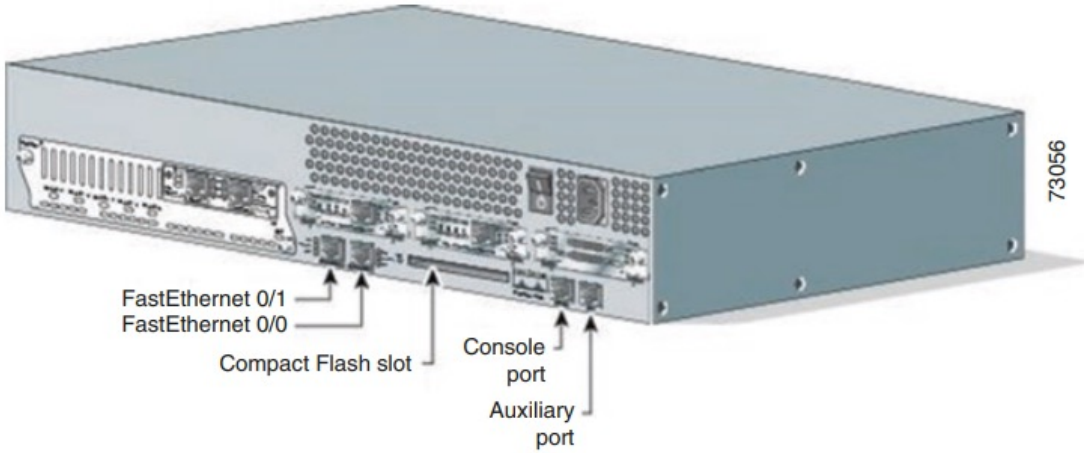


Router main components

NVRAM (Nonvolatile RAM)

Router Main Components

- ***Console (Management) Interface***: Console port or interface is the management port which is used by administrators to log on to a router directly (i.e., without using a network connection) via a computer with an RJ-45 or miniUSB connector.

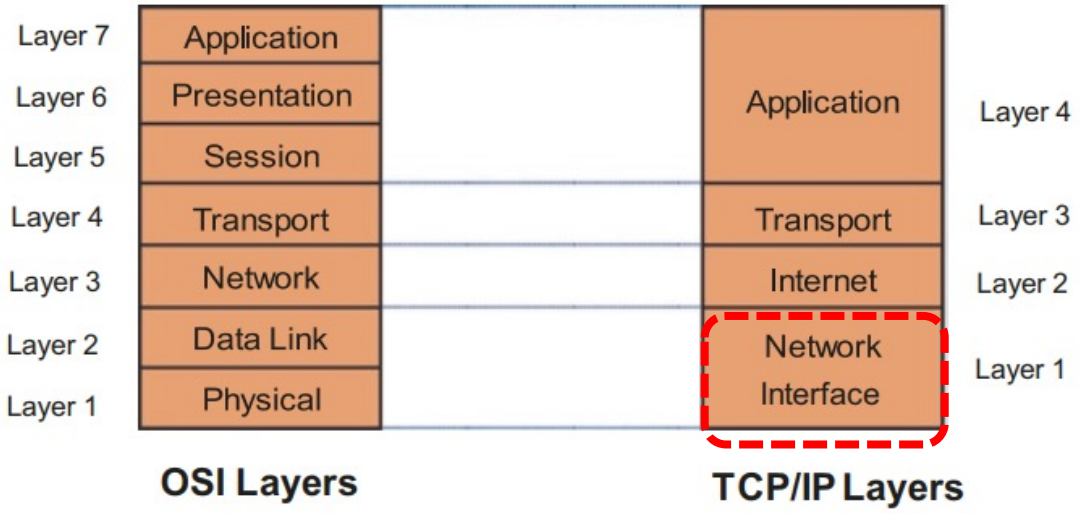


Example of a router's rear panel

3.2.Transmission Control Protocol/Internet Protocol (TCP/IP)

TCP/IP Layer 1: Network Access Layer

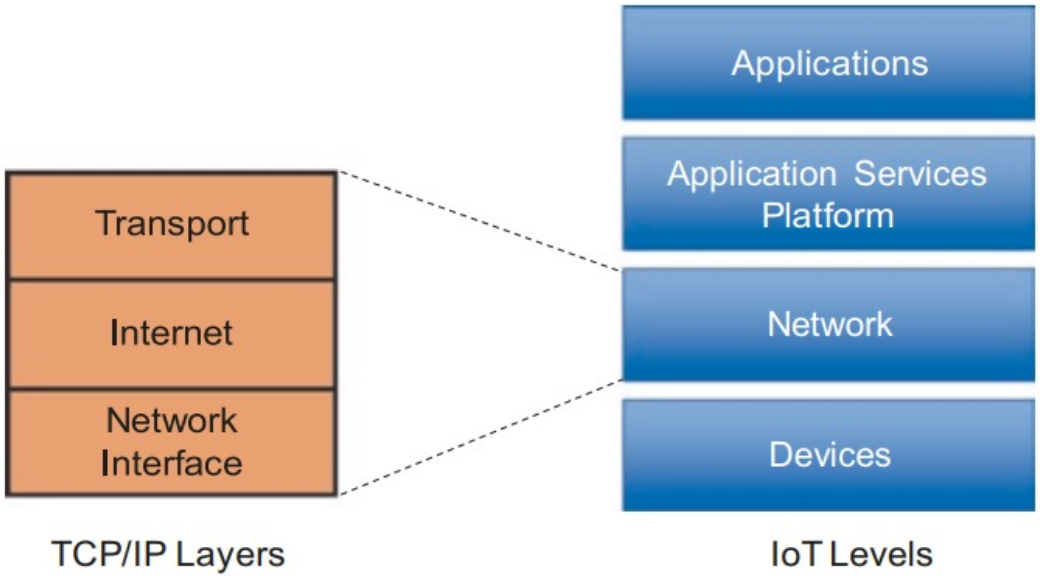
- **The Network Access Layer:** The Network Access Layer defines details of how data is physically sent through the network. This includes how bits are electrically or optically signaled by hardware devices that interface directly with a network medium, such as coaxial cable, optical fiber, radio links, or twisted pair copper wire.



Relationship between OSI reference model and TCP/IP

3.3. IoT Network Level: Key Performance Characteristics

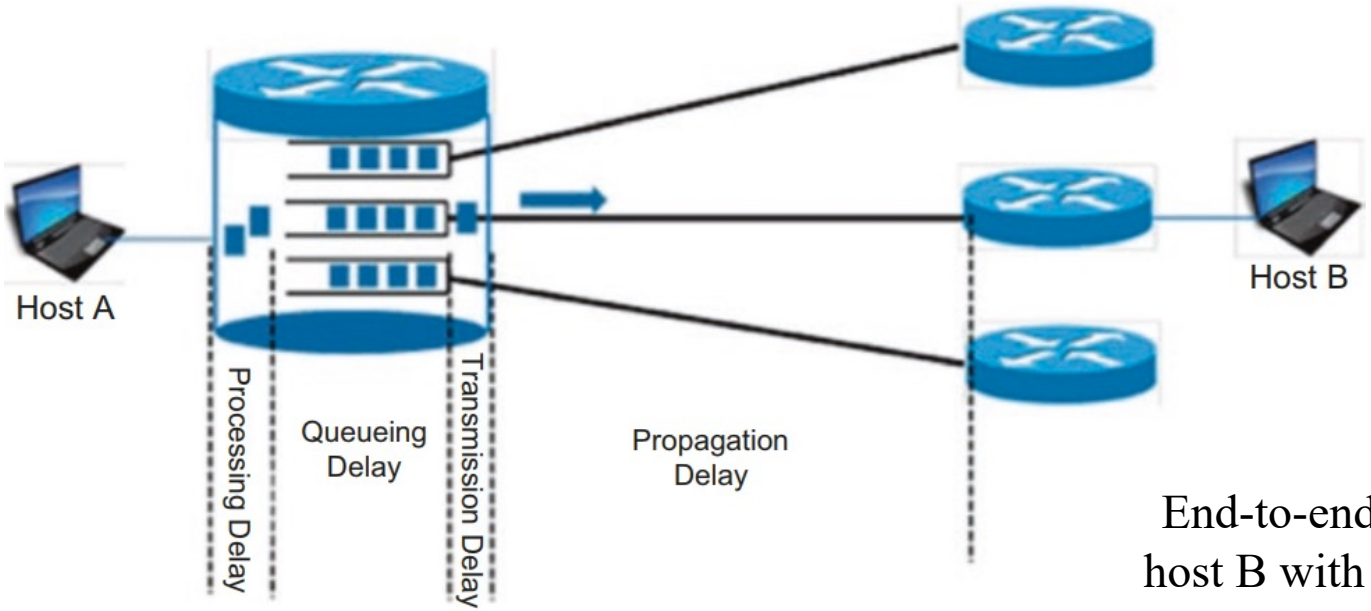
- The IoT Network Level is in fact the TCP/IP Layers. It should be noted that we have removed TCP/IP's Application Layer to prevent overlap with the IoT Application Level.



Mapping of IoT reference framework to TCP/IP layers

3.3. IoT Network Level: Key Performance Characteristics

- IoT Network Level key characteristics may be grouped into three main areas: end-to-end delay, packet loss, and network element throughput. Ideally, engineers want the IoT network to move data between any end points (or source and destination) instantaneously, without any delay or packet loss.



End-to-end delay from host A to host B with illustration at router A

3.3. IoT Network Level: Key Performance Characteristics

End-to-End Delay

End-to-end delay across the IoT network is perhaps the most essential performance characteristic for real-time applications especially in wide area networks (WAN) that connect multiple geographies

Processing delay: which is defined as the time a router takes to process the packet header and determine where to forward the packet. It may also include the time needed to check for bit-level errors in the packet (typically in the order of microseconds).

Queuing delay: which is defined as the time the packet spends in router queues as it awaits to be transmitted onto the outgoing link. Clearly Queuing delay depends on the number of earlier-arriving packets in the same queue (typically in the order of microseconds to milliseconds).

3.3. IoT Network Level: Key Performance Characteristics

End-to-End Delay

Transmission delay: which is defined as the time it takes to push the packet's bits onto the link. Transmission delay of packet of length L bits is defined L/R where R is the transmission rate of a link between two devices. For example, for a packet of length 1000 bits and a link of speed of 100 Mbps, the delay is 0.01 ms. (Transmission delay is typically in the order of microseconds to milliseconds.)

3.3. IoT Network Level: Key Performance Characteristics

End-to-End Delay

Propagation delay: which is defined as the time for a bit (of the packet) to propagate from the beginning of a link (once it leaves the source router) to reach its destination router. Hence, Propagation delay on a given link depends on the physical medium of the link itself (e.g., twisted pair copper, fiber, coaxial cable) and is equal to the distance of the link (between two routers) divided by the propagation speed (e.g., speed of light). (Propagation delay is typically in the order of milliseconds). It should be noted that unlike Transmission delay (i.e., the amount of time required to push a packet out), Propagation delay is independent of the packet length.

3.3. IoT Network Level: Key Performance Characteristics

End-to-End Delay

Hence, the total delay (d_{Total}), between two end points, is the sum of the Processing delay (d_{Process}), the Queuing delay (d_{Queue}), the Transmission delay (d_{Trans}), and the Propagation delay (d_{Prop}) across utilized network elements in the path, i.e

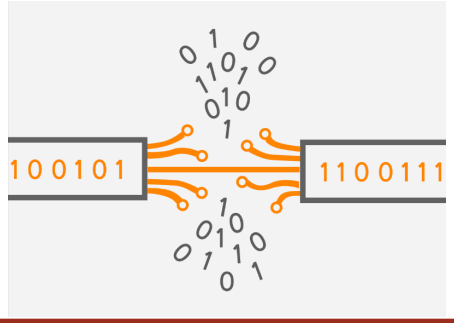
$$D_{\text{Total}} = d_{\text{Process}} + d_{\text{Queue}} + d_{\text{Trans}} + d_{\text{Prop}}$$

End-to-end delay is typically measured using Traceroute utility (available on many modern operating systems) as well as vendor-specific tools (e.g., Cisco's IP SLA (service-level agreement) that continuously collects data about delay, jitter, response time, and packet loss).

3.3. IoT Network Level: Key Performance Characteristics

Packet Loss

- Packet loss occurs when at least one packet of data traveling across a network fails to reach its destination.
- In general, packets are dropped and consequently lost when the network is congested (i.e., one of the network elements is already operating at full capacity and cannot keep up with arriving packets). This is due to the fact that both queues and links have finite capacities.
- Hence, a main reason for packet loss is link or queue congestion (i.e., a link between two devices, and its associated queues, is fully occupied when data arrives).



3.3. IoT Network Level: Key Performance Characteristics

Throughput

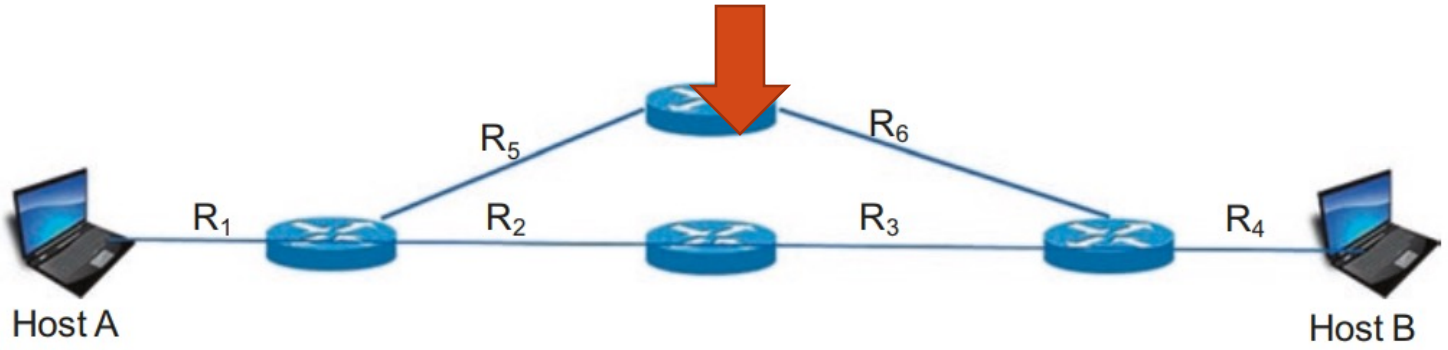
- Throughput may be defined as the maximum amount of data moved successfully between two end points in a given amount of time. Related measures include the link and device speed (how fast a link or a device can process the information) and response time (the amount of time to receive a response once the request is sent).
- Throughput is one of the key performance measures for network and computing devices and is typically measured in bits per second (bps) or gigabits per second (Gbps) at least for larger network devices. The system throughput is typically calculated by aggregating all throughputs across end points in a network (i.e., sum of successful data delivered to all destination terminals in a given amount of time).

3.3. IoT Network Level: Key Performance Characteristics

Throughput



$$\min\left(R_1, R_2, R_3 \text{ and } R_4\right).$$



3.4. Internet Protocol Suite

- As we mentioned earlier, TCP/IP provides end-to-end connectivity specifying how data should be packetized, addressed, transmitted, routed, and received at the destination.
- The objective of this chapter is not to provide an exhaustive list of the TCP/IP protocols but rather to provide a summary of the key protocols that are essential for IoT.

IoT Network Level: Addressing

IP addresses are globally unique numbers assigned by the Network Information Center. Globally unique addresses permit IP networks anywhere in the world to communicate with each other. Most of existing networks today use IP version 4 (IPv4). Advanced networks use IP version 6 (IPv6).

3.4. Internet Protocol Suite

IP Version 4

IPv4 addresses are normally expressed in dotted-decimal format, with four numbers separated by periods, such as 192.168.10.10. It consists of 4-octets (32-bit) number that uniquely identifies a specific TCP/IP (or IoT) network and a host (computer, printer, router, IP-enabled sensor, any device requiring a network interface card) within the identified network. IPv4 address consists of two main parts: the network address part and the host address part.

192	. 168	. 1	. 1	
8bits	8 bits	8 bits	8 bits =	32 bits
1 octet	1 octet	1 octet	1 octet =	4 octets

IP Version 4

- For example, the dotted-decimal IP address 192.168.10.8 is (in binary notation) the 32 bit number 11000000.10101000.000010 10.00001000. The decimal numbers separated by periods are the octets converted from binary to decimal notation.
- The first part of an IP address is used as a network address and the last part as a host address. If you take the example 192.168.10.8 and divide it into these two parts, you get the following: 192.168.10.0 network address and .8 host address or 192.168.10.0 network address and 0.0.0.8 host address.

IP Version 4

- In TCP/IP, the parts of the IP address that are used as the network and host addresses are not fixed, so the network and host addresses above cannot be determined unless you have more information. This information is supplied in another 32-bit number called a subnet mask. In the above example, the subnet mask is 255.255.255.0. It is not obvious what this number means unless you know that 255 in binary notation equals 11111111; so, the subnet mask is

11111111.11111111.11111111.0000000

IP Version 4 (IPv4)

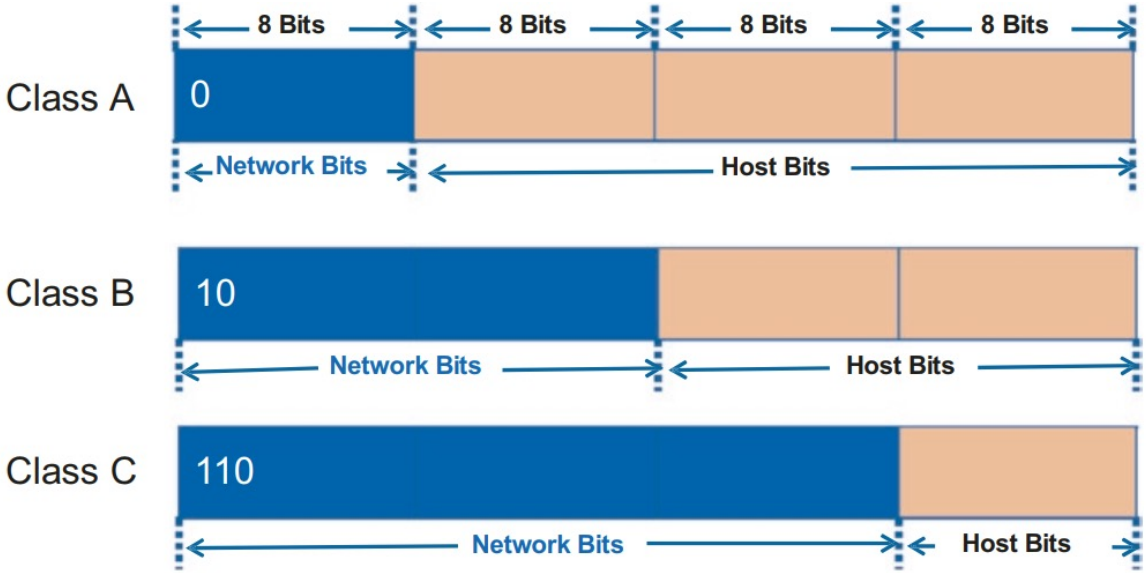
- Lining up the IP address and the subnet mask together, the network and host portions of the address can be separated:

$$11000000.10101000.00001010.10001000 - -\text{IP address}\,(192.168.10.8)$$

$$11111111.11111111.11111111.00000000 - -\text{Subnet mask}\,(255.255.255.0)$$

IP Version 4 (IPv4)

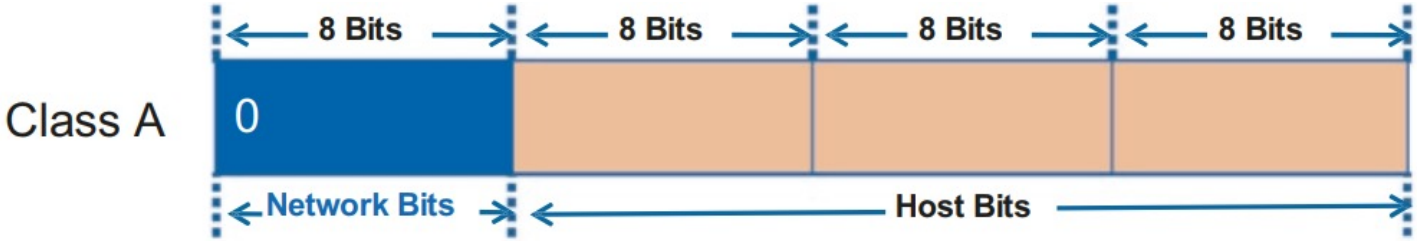
- Five classes (A, B, C, D, and E) have been established to identify the network and host parts. All the five classes are identified by the first octet of IP address. Classes A, B, and C are used in actual networks. Class D is reserved for multicasting (data is not destined for a particular host; hence there is no need to extract host address from the IP address). Class E is reserved for experimental purposes.



IPv4 address formats for classes A, B, and C

IP Version 4 (IPv4)

- Class A networks provide only 8 bits for the network address field and 24 bits for host address. It is intended mainly for use with very large networks with large number of hosts.
- The first bit of the first octet is always set to 0 (zero). Thus the first octet ranges from 1 to 127, i.e., 00000001–011111111. Class A addresses only include IP starting from 1.x.x.x to 126.x.x.x only. The IP range 127.x.x.x is reserved for loopback IP addresses. The default subnet mask for class A IP address is 255.0.0.0 which implies that class A addressing can have 126 networks (2^7–2) and 16,777,214 hosts (2^{24}–2).



IPv4 address formats for classes A

IP Version 4 (IPv4)

- Class B networks allocate 16 bits for the network address field and 16 bits for the host address filed. An IP address which belongs to class B has the first two bits in the first octet set to 10, i.e., 10000000–10111111 or 128–191 in decimal. Class B IP addresses range from 128.0.x.x to 191.255.x.x. The default subnet mask for class B is 255.255.x.x. Class B has 16,384 (2^{14}) network addresses and 65,534 (2^{16}–2) host addresses.



IPv4 address formats for classes B

IP Version 4 (IPv4)

- Class C networks allocate 24 bits for the network address field only 8 bits for the host field. Hence, the number of hosts per network may be a limiting factor. The first octet of Class C IP address has its first 3 bits set to 110, that is: 1110 0000–1110 1111 or 224–239 in decimal.
- Class C IP addresses range from 192.0.0.x to 223.255.255.x. The default subnet mask for Class C is 255.255.255.x. Class C gives 2,097,152 (2^{21}) Network addresses and 254 (2^8–2) Host addresses.



IPv4 address formats for classes C

IP Version 6 (IPv6)

- IPv4 has room for about 4.3 billion addresses, which is not nearly enough for the world's people, let alone IoT with a forecast of 20 billion devices by 2020.

IP Version 6 (IPv6)

- The IPv6 address is 128 bits long. It is divided into blocks of 16 bits. Each 16-bit block is then converted to a 4-digit hexadecimal number, separated by colons. The resulting representation is called colon-hexadecimal. This is in contrast to the 32-bit IPv4 address represented in dotted-decimal format, divided along 8-bit boundaries, and then converted to its decimal equivalent, separated by periods

(a) *Binary Form*

- 01110001110110100000000011010011000000000000000001011110011011
- 00000010101010100000000011111111111111110001010001001110001011011

IP Version 6 (IPv6)

(b) *16-Bit Boundaries Form*

- 0111000111011010 0000000011010011 0000000000000000 0010111100111011
- 0000001010101010 0000000011111111 1111111000101000 1001110001011011

(c) 16-Bit Block Hexadecimal *and Delimited with Colons Form*

- 71DA:00D3:0000:2F3B:02AA:00FF:FE28:9C5B.
- i.e., $(0111000111011010)_2 = (71DA)_{16}$, $(0000000011010011)_2 = (D3)_{16}$, and so on.

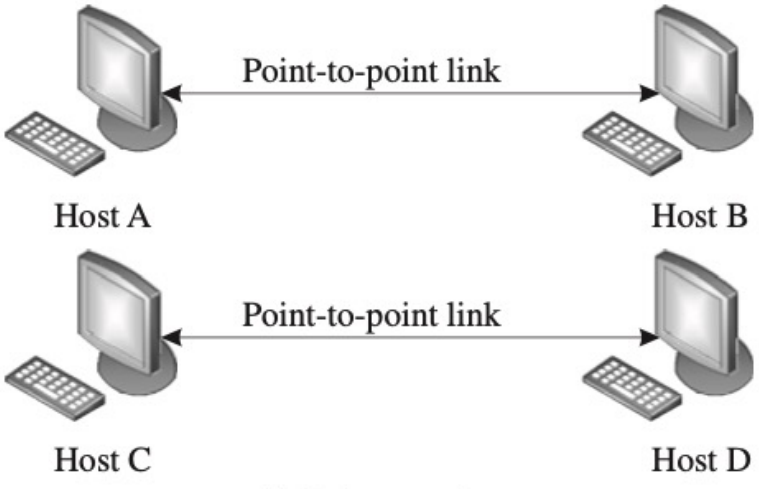
(d) *Final Form (16-Bit Block Hexadecimal and Delimited with Colons Form, Simplified by Removing the Leading Zeros).*

- 71DA:D3:0:2F3B:2AA:FF:FE28:9C5B

3.5. Network Types

Connection types

Point-to-point: Point-to-point connections are used to establish direct connections between two hosts. Day-to-day systems such as a remote control for an air conditioner or television is a point to point connection, where the connection has the whole channel dedicated to it only. These networks were designed to work over duplex links and are functional for both synchronous as well as asynchronous systems.

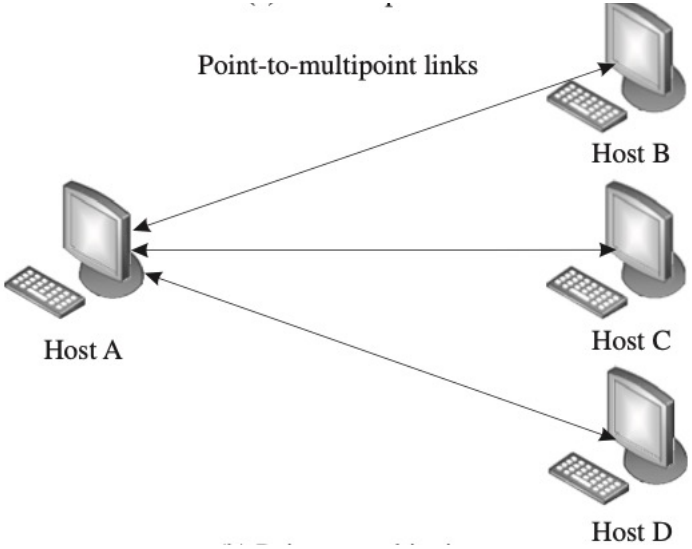


Point-to-point

3.5. Network Types

Connection types

Point-to-multipoint: In a point-to-multipoint connection, more than two hosts share the same link. This type of configuration is similar to the one-to-many connection type. Point-to-multipoint connections find popular use in wireless networks and IP telephony. The channel is shared between the various hosts, either spatially or temporally.

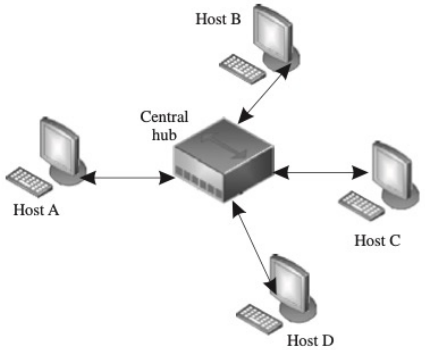


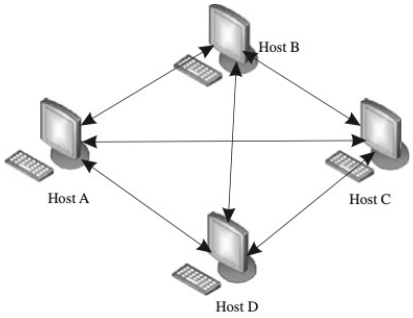
Point-to-multipoint

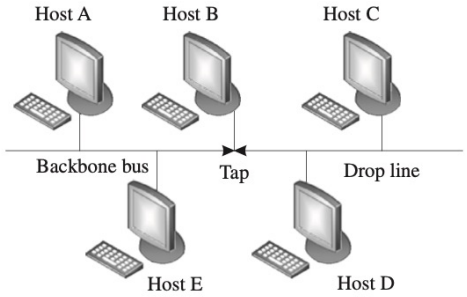
3.5. Network Types

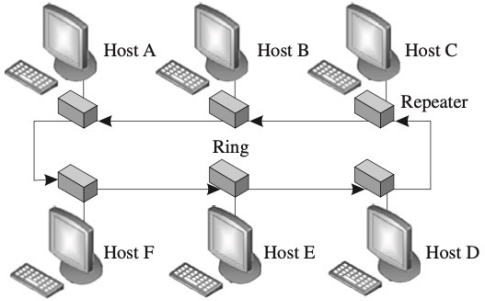
Physical topology

Depending on the physical manner in which communication paths between the hosts are connected, computer networks can have the following four broad topologies: *Star*, *Mesh*, *Bus*, and *Ring*.





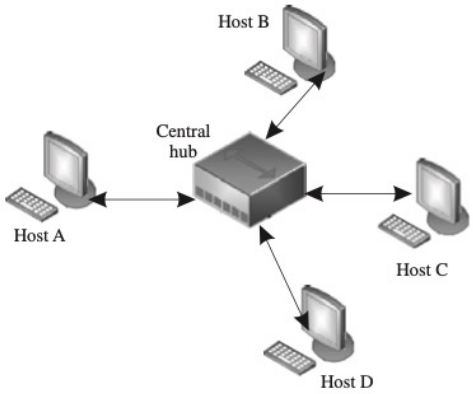




3.5. Network Types

Star

In a star topology, every host has a point-to-point link to a central controller or hub. The hosts cannot communicate with one another directly; they can only do so through the central hub. The hub acts as the network traffic exchange. For large-scale systems, the hub, essentially, has to be a powerful server to handle all the simultaneous traffic flowing through it. The main advantages of the star topology are easy installation and the ease of fault identification within the network. However, the main disadvantage of this topology is the danger of a single point of failure. If the hub fails, the whole network fails.



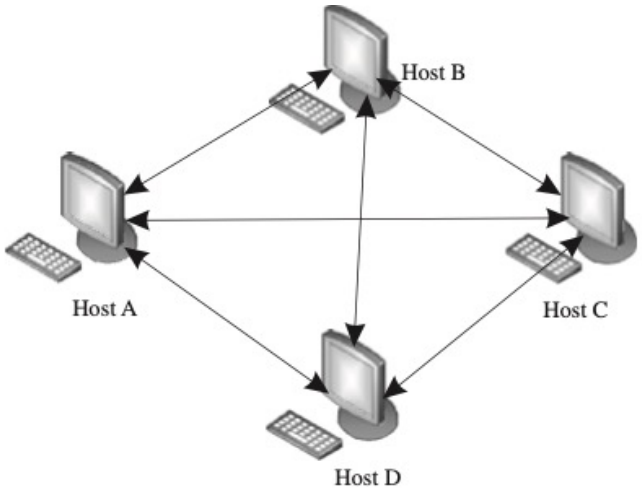
Star topology

3.5. Network Types

Mesh

In a mesh topology, every host is connected to every other host is connected to every other host using a dedicated link (in a point-to-point manner).

The first significant advantage is the robustness and resilience of the system. Even if a link is down or broken, the network is still fully functional as there remain other pathways for the traffic to flow through. The second advantage is the security and privacy of the traffic as the data is only seen by the intended recipients and not by all members of the network.

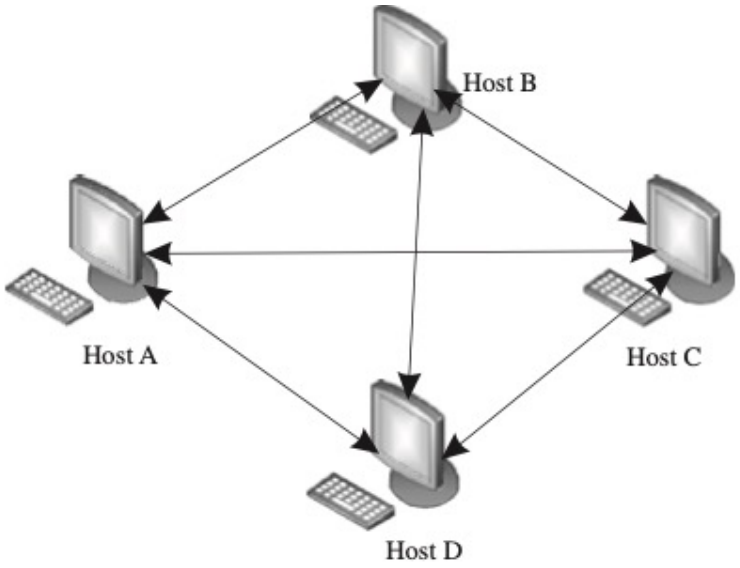


Mesh topology

3.5. Network Types

Mesh

The third advantage is the reduced data load on a single host, as every host in this network takes care of its traffic load. However, owing to the complexities in forming physical connection between devices and the cost of establishing these links, mesh networks are used very selectively, such as in backbone networks.

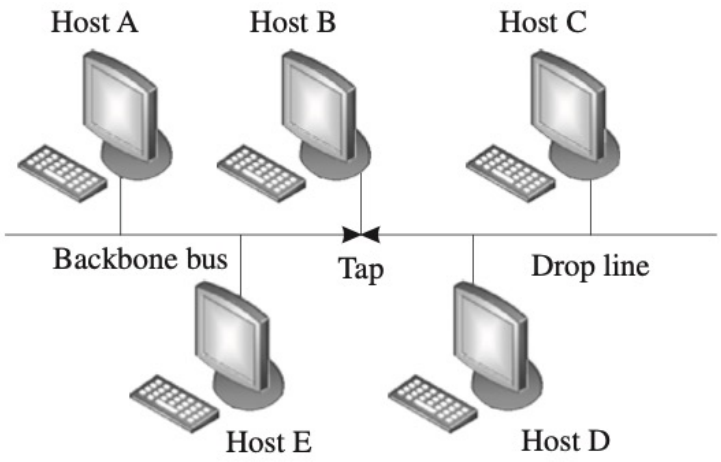


Mesh topology

3.5. Network Types

Bus

A bus topology follows the point-to-multipoint connection. A backbone cable or bus serves as the primary traffic pathway between the hosts. The hosts are connected to the main bus employing drop lines or taps. The main advantage of this topology is the ease of installation. However, there is a restriction on the length of the bus and the number of hosts that can be simultaneously connected to the bus due to signal loss over the extended bus.

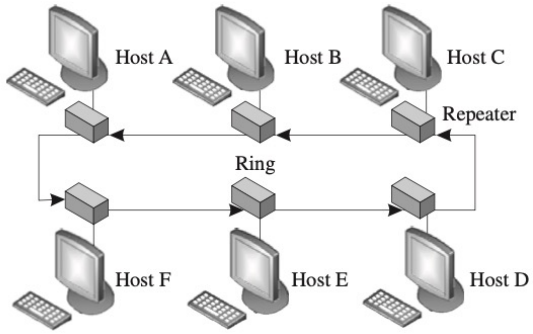


Bus topology

3.5. Network Types

Bus

A ring topology works on the principle of a point-to-point connection. Here, each host is configured to have a dedicated point-to-point connection with its two immediate neighboring hosts on either side of it through repeaters at each host. The repetition of this system forms a ring. The repeaters at each host capture the incoming signal intended for other hosts, regenerates the bit stream, and passes it onto the next repeater. Fault identification and set up of the ring topology is quite simple and straightforward. However, the main disadvantage of this system is the high probability of a single point of failure. If even one repeater fails, the whole network goes down.



Ring topology

3.5. Network Types

Physical topology

Network topology comparison

Topology	Feature	Advantage	Disadvantage
Star	Point-to-point	Cheap; ease of installation; ease of fault identification	Single point of failure; traffic visible to network entities
Mesh	Point-to-point	Resilient against single point of failures; scalable; traffic privacy and security ensured	Costly; complex connections
Bus	Point-to-multipoint	Ease of installation; cheap	Length of backbone cable limited; number of hosts limited; hard to localize faults
Ring	Point-to-point	Ease of installation; cheap; ease of fault identification	Prone to single point of failure

3.6. Network reachability

Personal Area Networks (PAN)

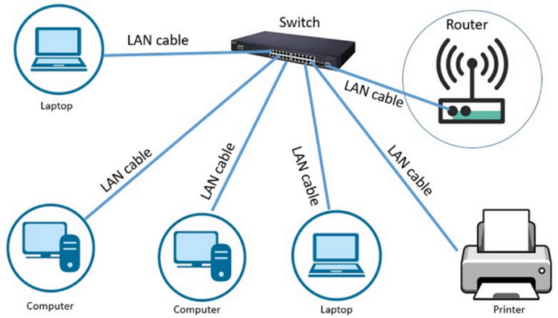
PANs, as the name suggests, are mostly restricted to individual usage. A good example of PANs may be connected wireless headphones, wireless speakers, laptops, smartphones, wireless keyboards, wireless mouse, and printers within a house. Generally, PANs are wireless networks, which make use of low-range and low-power technologies such as Bluetooth. The reachability of PANs lies in the range of a few centimeters to a few meters.



3.6. Network reachability

Local Area Networks (LAN)

A LAN is a collection of hosts linked to a single network through wired or wireless connections. However, LANs are restricted to buildings, organizations, or campuses. Typically, a few leased lines connected to the Internet provide web access to the whole organization or a campus; the lines are further redistributed to multiple hosts within the LAN enabling hosts. Typically, the present-day data access rates within the LANs range from 100 Mbps to 1000 Mbps, with very high fault-tolerance levels. Commonly used network components in a LAN are servers, hubs, routers, switches, terminals, and computers.



3.6. Network reachability

Metropolitan Area Networks (MAN)

The reachability of a MAN lies between that of a LAN and a WAN. Typically, MANs connect various organizations or buildings within a given geographic location or city. An excellent example of a MAN is an Internet service provider (ISP) supplying Internet connectivity to various organizations within a city. As MANs are costly, they may not be owned by individuals or even single organizations. Typical networking devices/components in MANs are modems and cables. MANs tend to have moderate fault tolerance levels.

3.6. Network reachability

Wide Area Networks (WAN)

WANs typically connect diverse geographic locations. However, they are restricted within the boundaries of a state or country. The data rate of WANs is in the order of a fraction of LAN's data rate. Typically, WANs connecting two LANs or MANs may use public switched telephone networks (PSTNs) or satellite-based links. Due to the long transmission ranges, WANs tend to have more errors and noise during transmission and are very costly to maintain. The fault tolerance of WANs are also generally low.

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

- Để triển khai được các kiến trúc truyền tải dữ liệu, các nền tảng phần mềm, hệ điều hành chạy trên thiết bị IoT phải hỗ trợ các chuẩn kết nối không dây mà ở đó những cơ chế, giải thuật định tuyến, điều khiển kết nối giữa các thiết bị hoạt động đúng như thiết kế. Tập hợp tất cả cơ chế, giải thuật, quy trình đó được định nghĩa và thiết kế theo một tiêu chuẩn mạng, tạo nên các bộ giao thức mạng đảm bảo chức năng truyền gửi dữ liệu và tín hiệu điều khiển trong mạng cảm biến không dây cũng như mạng Internet bên ngoài.

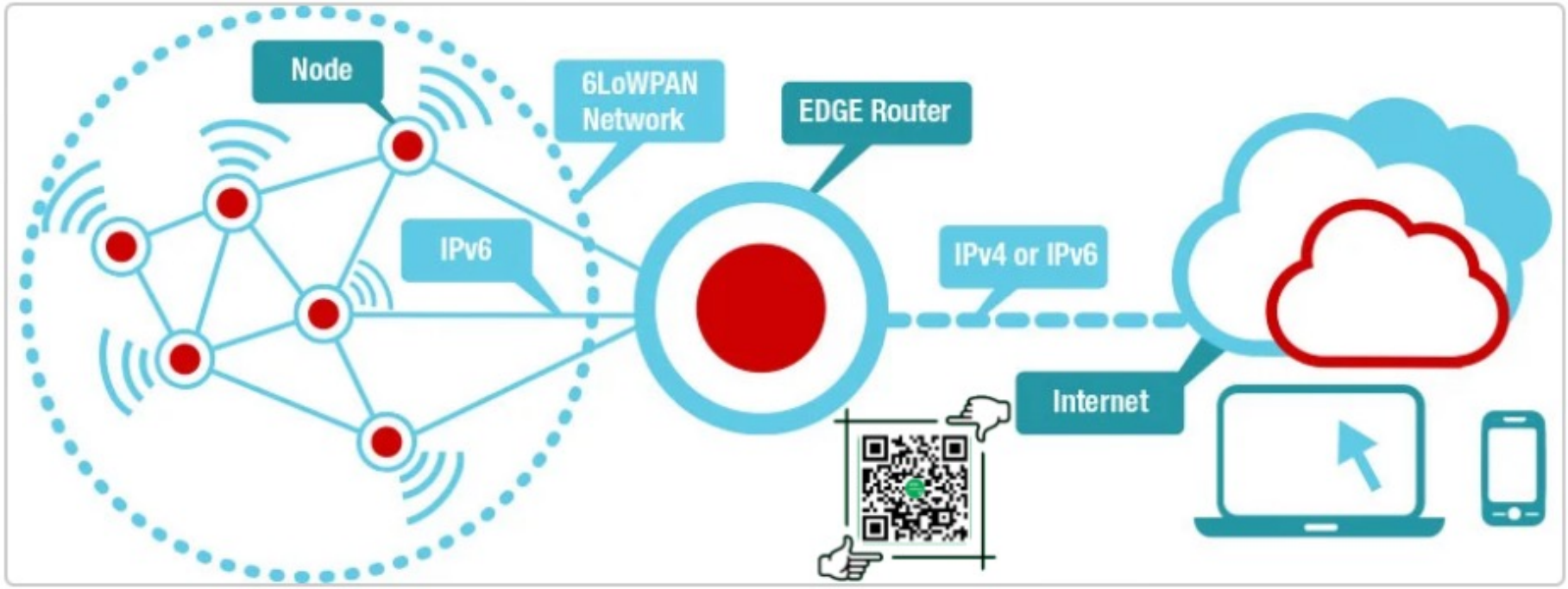
3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Bộ giao thức 6LowPAN

- 6LowPAN là viết tắt của "**I**Pv6 over **Lo**w-**P**ower **W**ireless **P**ersonal **A**rea **N**etwork", là công nghệ mạng cho phép gói tin IPv6 được sử dung trong những link layer nhỏ như định nghĩa trong *IEEE 802.15.4*. Là lựa chọn tốt cho các ứng dung IoT khi mạng cảm biến không dây có kích thước lớn.

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Kiến trúc mạng 6LoWPAN



3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Kiến trúc mạng 6LoWPAN

- Có hai loại thiết bị phổ biến bên trong mạng 6LoWPAN là: router và host; trong đó router dùng để định tuyến đường đi tới các node trong mạng 6LoWPAN, host chính là các thiết bị cuối, không có khả năng định tuyến., host có thể hoạt động ở chế độ "ngủ" và được đánh thức định kỳ để tiết kiệm năng lượng tiêu thụ.

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Nền tảng hiện thực bộ giao thức 6LoWPAN

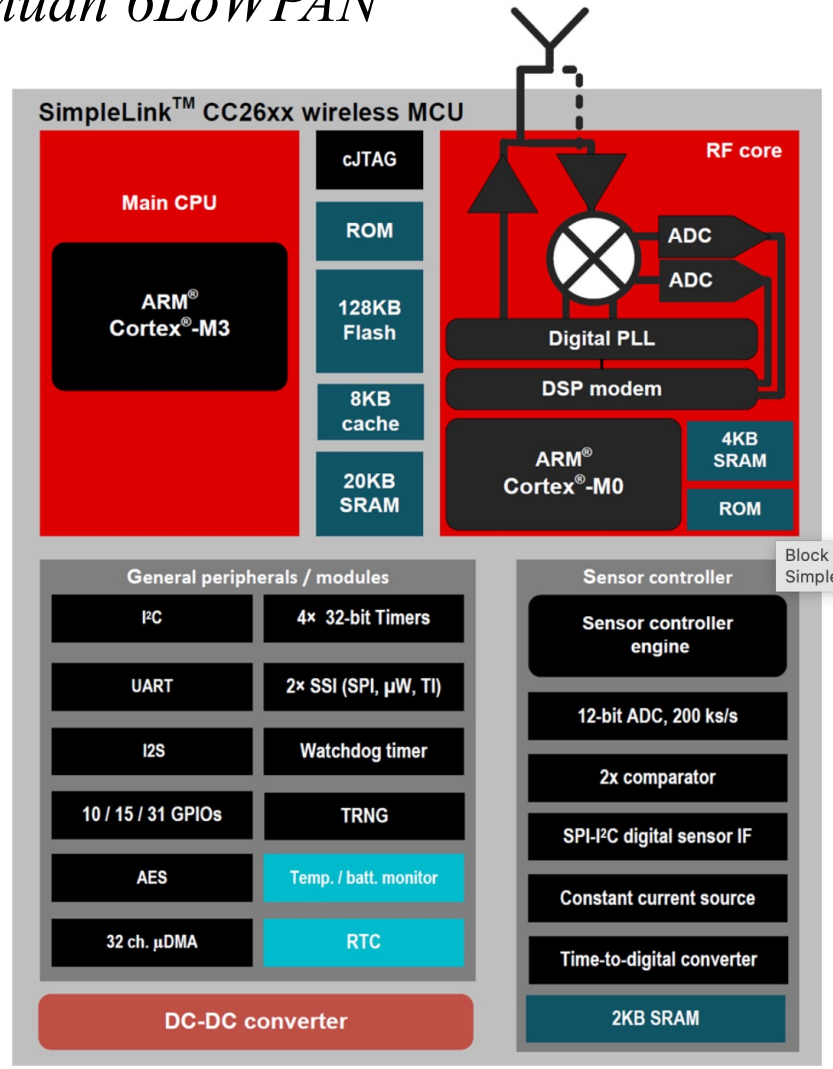
- Bộ giao thức mạng 6LoWPAN là một chuẩn giao thức được sử dung phổ biến trong nhiều giải pháp IoT.
- Với đặc trưng cho phép thiết bị IoT giao tiếp trực tiếp với mạng Internet bên ngoài thông qua địa chỉ IPv6, giao thức 6LoWPAN đã được nghiên cứu và hiện thực trên nhiều nền tảng phần cứng, phần mềm khác nhau.

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Một số nền tảng phần cứng hỗ trợ chuẩn 6LoWPAN

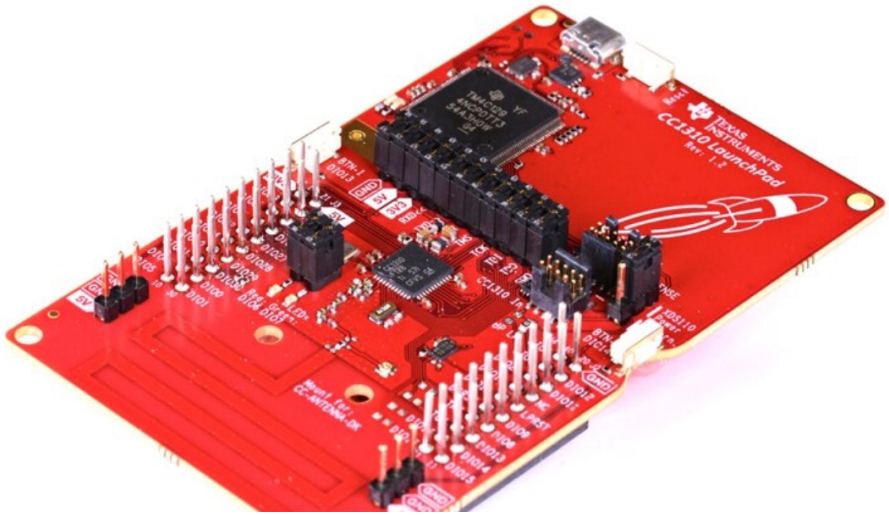


CC2650 SimpleLink 6LoWPAN/ZigBee Wireless MCUs



3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Một số nền tảng phần cứng hỗ trợ chuẩn 6LoWPAN



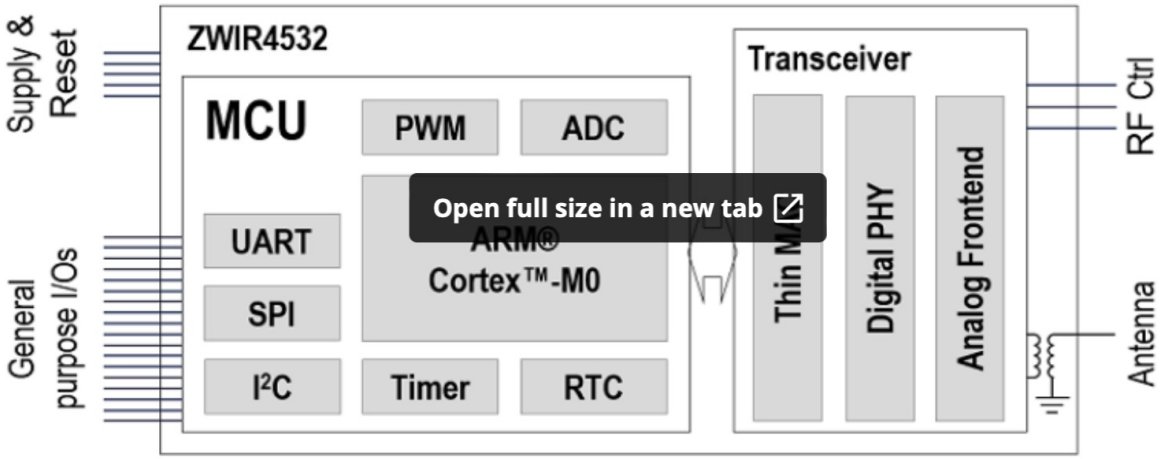


CC1310 SimpleLink Wireless MCUs

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Một số nền tảng phần cứng hỗ trợ chuẩn 6LoWPAN



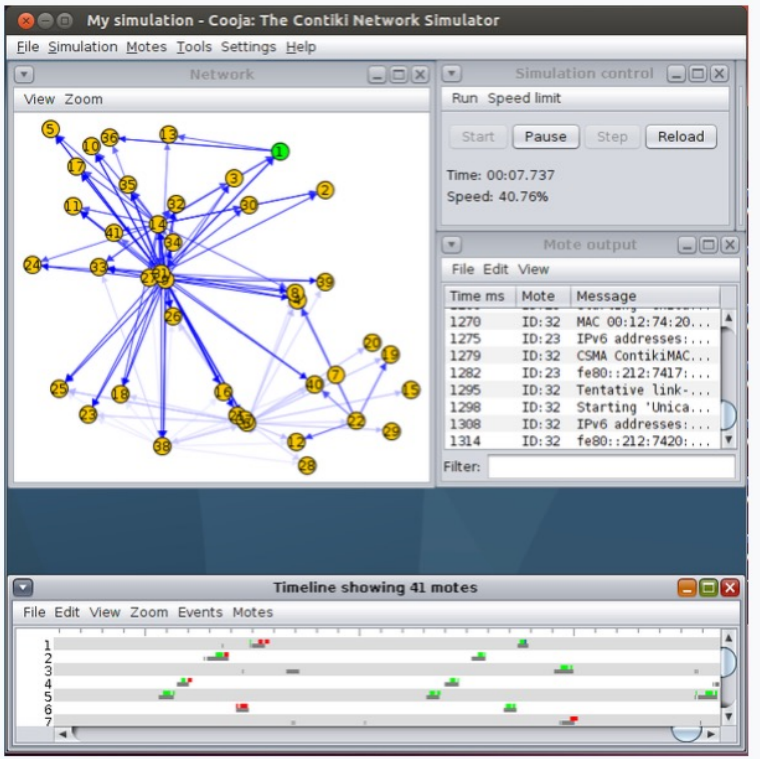


ZWIR4532 - Block Diagram

ZWIR4532 IoT 6LoWPAN Module

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Một số nền tảng phần mềm hỗ trợ bộ giao thức 6LoWPAN



Contiki

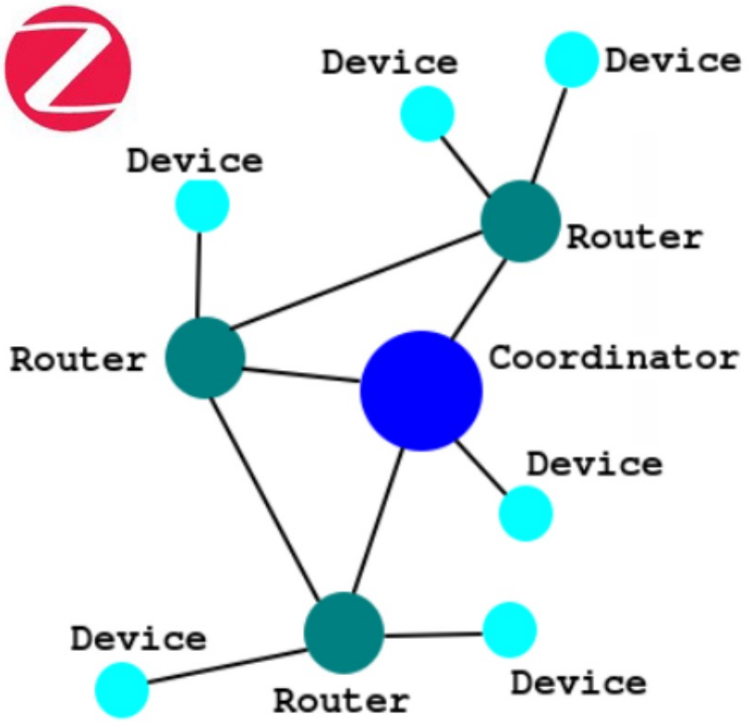
Mbed OS

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Bộ giao thức Zigbee

Mạng Zigbee có 3 nhóm thiết bị logic (theo chức năng):

- ❖ Thiết bị điều phối mạng (Coordinator).
- ❖ Bộ định tuyến (Router).
- ❖ Thiết bị đầu cuối.



Mô hình truyền dẫn mạng Zigbee

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Kiến trúc mạng Zigbee

Coordinator:

Được sử dụng để tạo mạng Zigbee. Khi một nút kết nối, nó sẽ cấp cho địa chỉ cho nút con. Mỗi mạng Zigbee cần và chỉ cần một Coordinator. Các mạng khác nhau phải có PAN ID khác nhau. Nếu có hai Coordinator có cùng PAN ID thì Coordinator khởi động sau sẽ tự động them một vào PAN ID để tránh xung đột.

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Kiến trúc mạng Zigbee

Router:

Chịu trách nhiệm chuyển tiếp các gói dữ liệu để tìm đường truyền thích hợp nhất. Mỗi mạng Zigbee có thể yêu cầu nhiều hơn 1 Router. Mỗi Router không những có thể gửi hoặc nhận dữ liệu mà còn có thể chuyển tiếp dữ lieu. Cấu trúc mạng Zigbee:

- Mạng Zigbee chứa 1 Coordinator và *n* Router, tất cả chúng đều phải có chung PAN ID.
- Mỗi nút có thể gửi và nhận dữ liệu, cũng có thể chuyển tiếp. Mỗi nút có thể kết nối tối đa đến 6 nút con.

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Kiến trúc mạng Zigbee

Router:

- Router được nhận một địa chỉ (địa chỉ ngắn – Short Address) là hằng số, có thể được sử dụng trong truyền điểm – điểm.
- Kể cả khi Coordinator bị ngắt điện, Router vẫn có thể duy trì mạng lưới, vì vậy, giữa Router và Router vẫn có thể giao tiếp được.

Khi một nút được thêm vào ngay cả khi Coordinator bị ngắt điện, nó vẫn có thể thông qua các địa chỉ của Router hiện hữu để tham gia vào mạng lưới. Mô tả về 1 mạng Zigbee bao gồm 1 Coordinator, 6 Router, và một số End Devices.

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Một số đặc điểm của chuẩn Zigbee trong ứng dung mạng cảm biến không dây

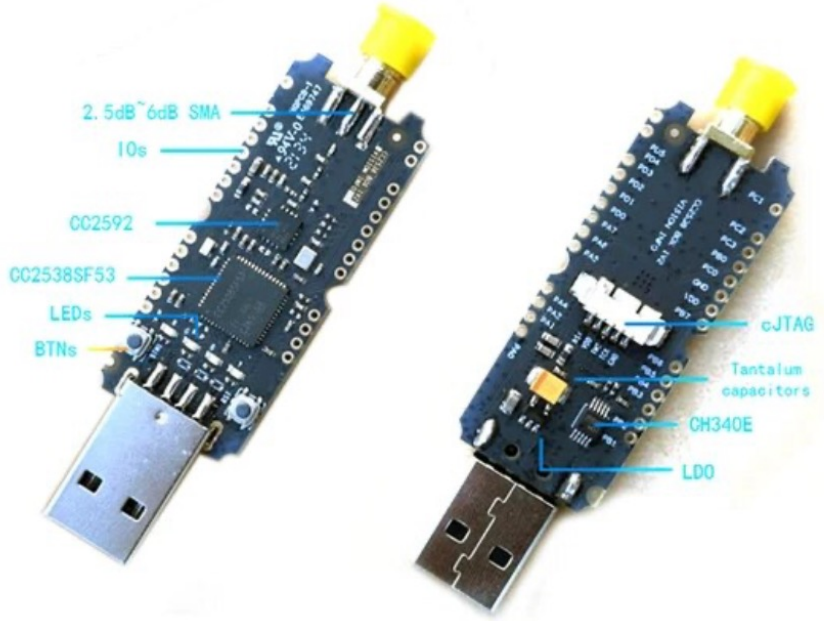
- Zigbee có độ tin cậy cao.
- Zigbee có chi phí thấp.
- Zigbee sử dụng năng lượng thấp.
- Zigbee bảo mật cao.
- Zigbee là một chuẩn mở toàn bộ.
- Zigbee có tốc độ truyền dữ liệu thấp.
- Zigbee bảo mật cao.
- Zigbee là một chuẩn mở toàn bộ.
- Zigbee có tốc độ dữ liệu thấp.

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Nền tảng hiện thực bộ giao thức mạng Zigbee



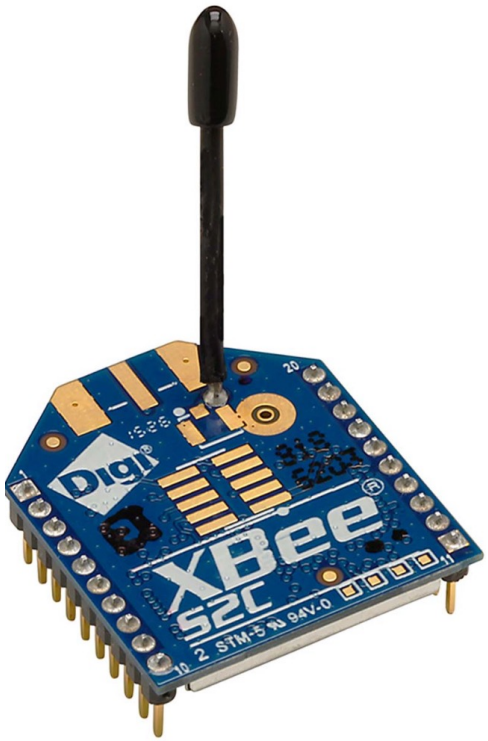
CC2530



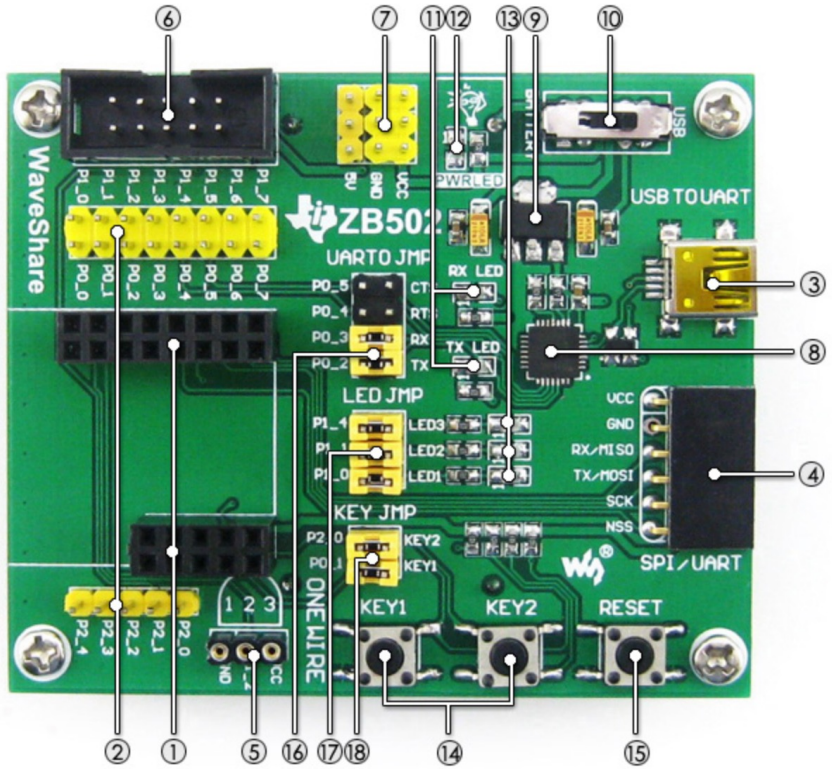
CC2538

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Nền tảng hiện thực bộ giao thức mạng Zigbee



Digi Xbee Zigbee



CC2530 Module Connector

3.7.Các bộ giao thức mạng không dây thường dùng trong IoT

Nền tảng hiện thực bộ giao thức mạng Zigbee



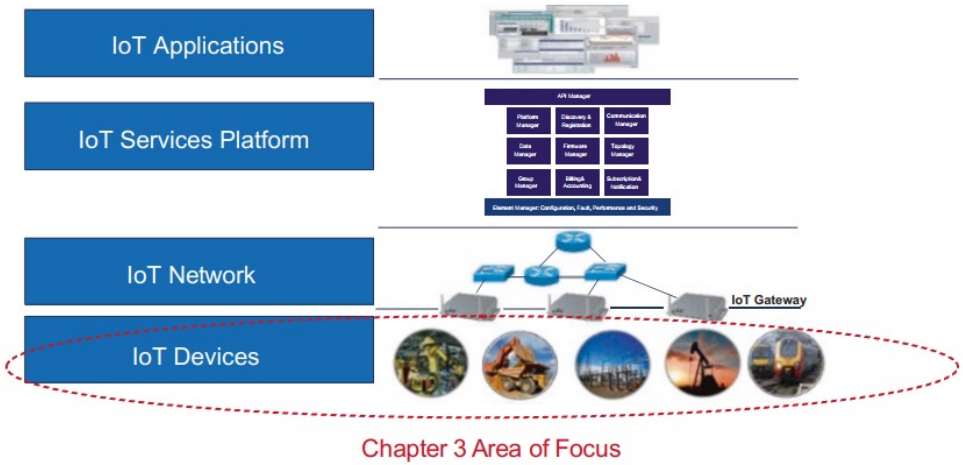
Arduino-based



nRF5 SDK

3.8. IoT Sensors

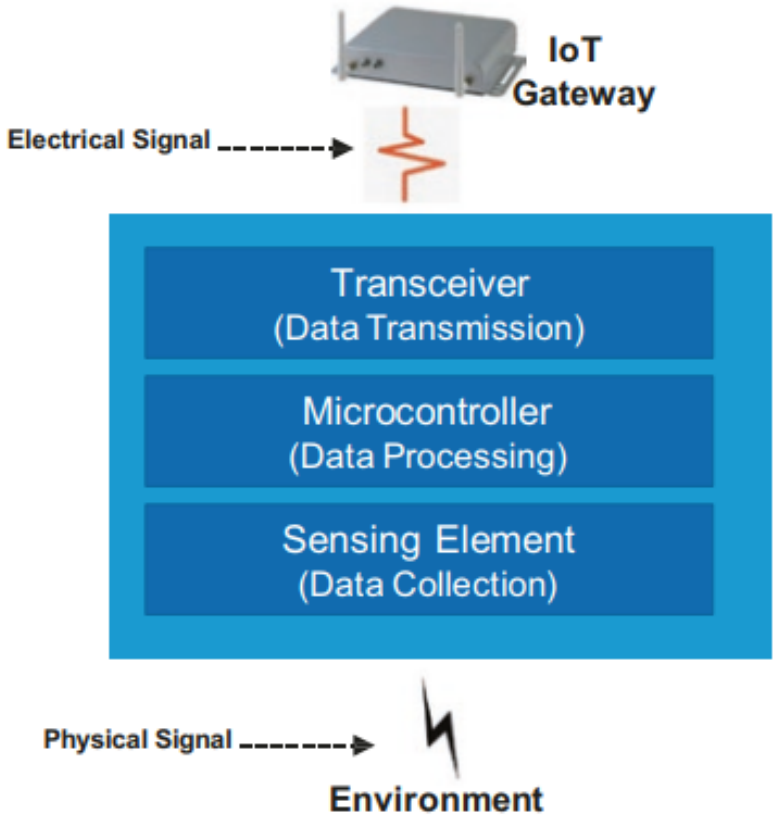




3.8. IoT Sensors

Definition

- A sensor is a device (typically electronic) that detects events or changes in its physical environment (e.g., temperature, sound, heat, pressure, flow, magnetism, motion, chemical and biochemical parameters) and provides a corresponding output.
- Most sensors take analog inputs and deliver digital, often electrical, outputs.



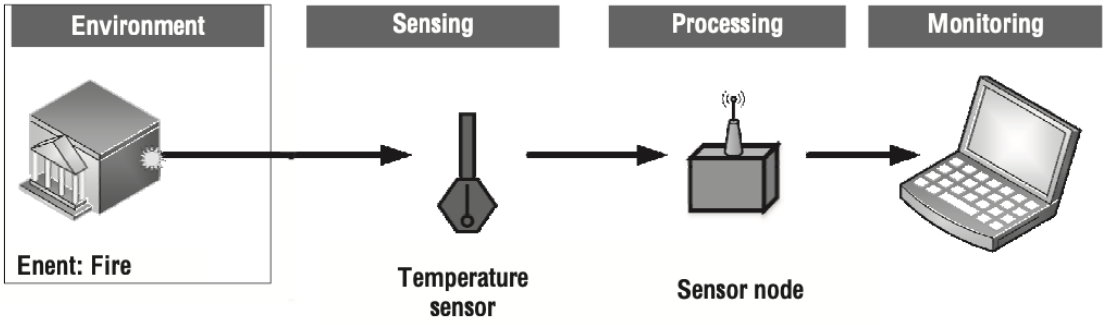
Components of smart sensors

3.8. IoT Sensors

Why Sensors?

As we mentioned above, a sensor's main purpose is collecting data from its surrounding environment and providing output to its adjoining devices (e.g., gateways, actuators) or applications.

Sensors typically collect data using physical interfaces (inputs) that sense the environment and then convert input signals into electrical signals (outputs) that are understood by the communication and computing devices. Output signals are then processed by the gateways and/or by applications of the IoT Platform.



The outline of a simple sensing operation

3.8. IoT Sensors

Sensor Types

Temperature Sensors: Temperature is perhaps the most commonly measured conservational quantity. This is anticipated since most physical, electronic, chemical, mechanical, and biological systems are affected by temperature.





3.8. IoT Sensors

Sensor Types

Pressure Sensors: Pressure sensors are used to measure the pressure of gases or liquids including water level, flow, speed, and altitude. Practical examples include sensors for pumps and compressors, hydraulic systems, and refrigerators.



3.8. IoT Sensors

Sensor Types

Flow Sensors: Flow sensors are used to detect and record the rate of fluid flow in a pipe or a system. They are also used to measure the flow/transfer of heat caused by the moving medium. Sensing and measuring the flow are critical for many applications ranging from bereave machine to more serious applications such as flow monitoring for high-purity acids.



3.8. IoT Sensors

Sensor Types

Level Sensors: Level sensors are used to measure the level of fluids continuously or at point values. The element to be measured can be inside a container or can be in its natural form such as a well in an oil rig.

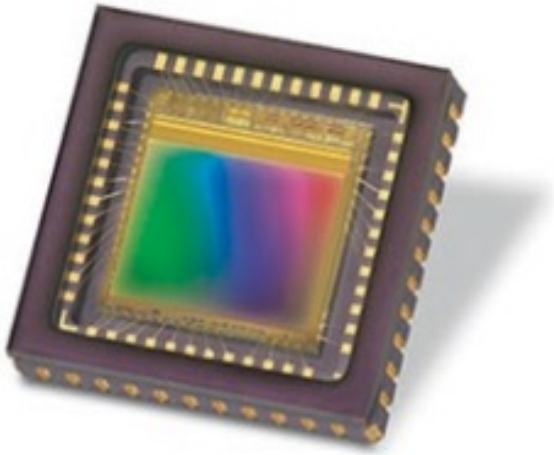




3.8. IoT Sensors

Sensor Types

Imaging Sensors: Imaging sensors are sophisticated sensors used in digital cameras, medical imaging machines, and night vision equipment. They are utilized to measure image information by capturing and then converting variable attenuation of waves into signals.





3.8. IoT Sensors

Sensor Types

Noise Sensors: High noise can have damaging effects on humans (e.g., cardiovascular) as well as animals (e.g., hearing loss). Such noise is often caused by machines, airplanes, trains, construction, and loud music especially in closed spaces.

Air Pollution Sensors: Many governments have established agencies to monitor and control the air quality in major cities. For instance, the USA has established the EPA (Environmental Protection Agency), in 1970, with a mission to protect Americans from significant health risks by providing accurate environmental information to its citizens.

3.8. IoT Sensors

Sensor Types

Infrared Sensors: Infrared sensors are used to track an object's movement. They produce and receive infrared waves in the form of heat.

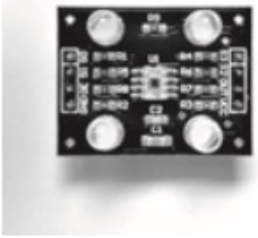
Moisture and Humidity Sensors: Moisture and humidity sensors (sometimes referred to as hygrometer sensors) are used to measure and report the relative humidity in the air. They use capacitive measurement by relying on electrical capacitance.

Speed Sensors: Speed sensors are commonly used to detect the speed of transport vehicles. Examples include wheel speed sensors, speedometers, Doppler radar, and laser surface velocimeter.

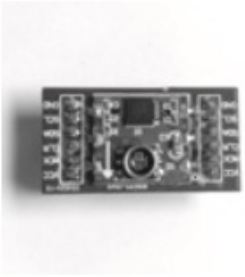
3.8. IoT Sensors



(a) Camera sensor



(b) Color sensor



(c) Compass and barometer



(d) Current sensor



(e) Digital temperature and humidity sensor



(f) Flame sensor



(g) Gas sensor



(h) Infrared sensor



(i) Rainfall sensor



(j) Ultrasonic distance measurement sensor

Some common commercially available sensors used for IoT-based sensing applications

3.9. RFID

- Another way of capturing information from “Things” is through the use of RFID (radio-frequency identification). RFID is not a sensor but a mechanism to capture information pre-embedded into the so-called Tag of a thing or an object using radio waves.
- RFID consists of two parts: a tag and a reader. Further, the tag has two parts: a microchip that stores and processes information and an antenna to receive and transmit a signal. The tag contains the specific serial number for one specific object. The reader reads the information encoded on a tag, using a two-way radio transmitterreceiver, by emitting a signal to the tag using an antenna.

3.9. RFID

An RFID-based solution has some advantages over older reader-tag-based solutions, such as barcode, including:

- RFID tag does not need to be within direct line of sight of the reader and can be read from a distance up to 12 m for passive ultrahigh frequency (UHF) system. Battery-powered tags typically have a reading range of 100 m.
- RFID data on the tag can be modified based on business needs. The barcode data is very difficult to change once deployed.
- RFID tags are durable. Barcodes, in comparison, are printed on a product for everyone to see. They can be damaged or changed. RFID tags are hidden and may be reused across multiple products. Also RFID tags are capable of storing much more data.

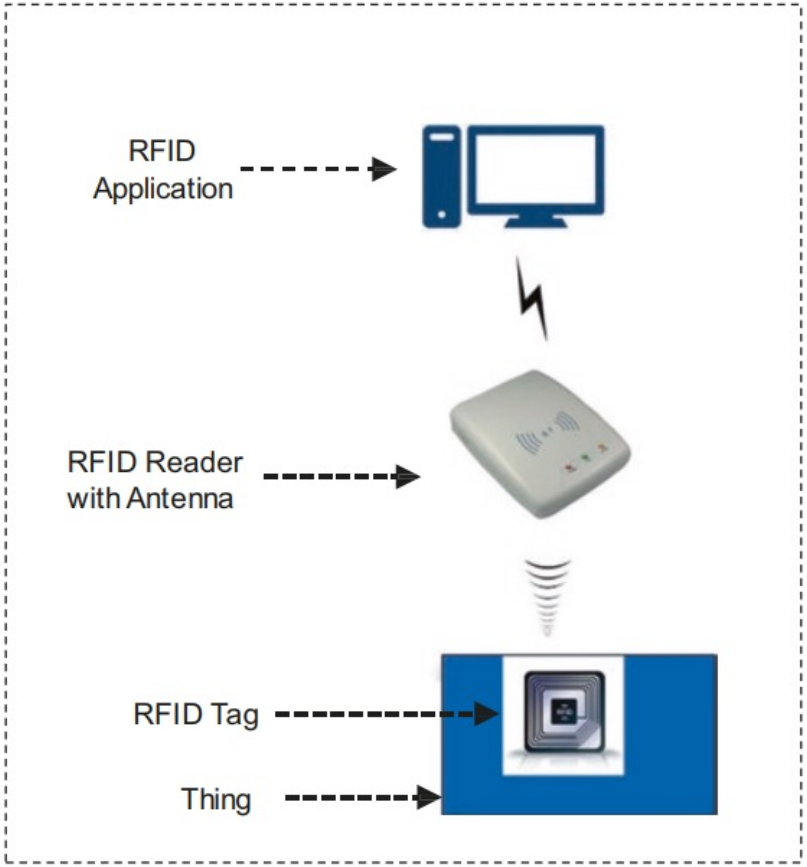
3.9. RFID

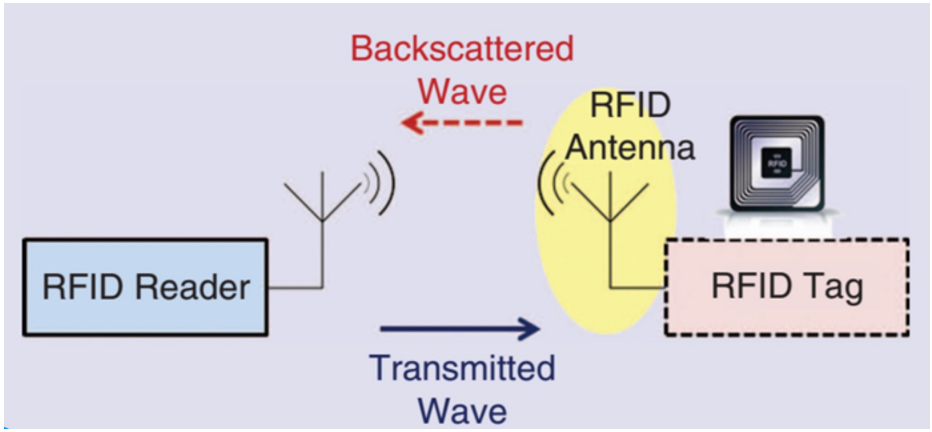
An RFID-based solution has some advantages over older reader-tag-based solutions, such as barcode, including:

- RFID data may be encrypted on the tag, thereby preventing unauthorized users from changing the data or counterfeiting.
- RFID systems can read hundreds of tags simultaneously. This is significant in a retail store as it saves the staff valuable time that they can spend on higher value tasks.

3.9. RFID

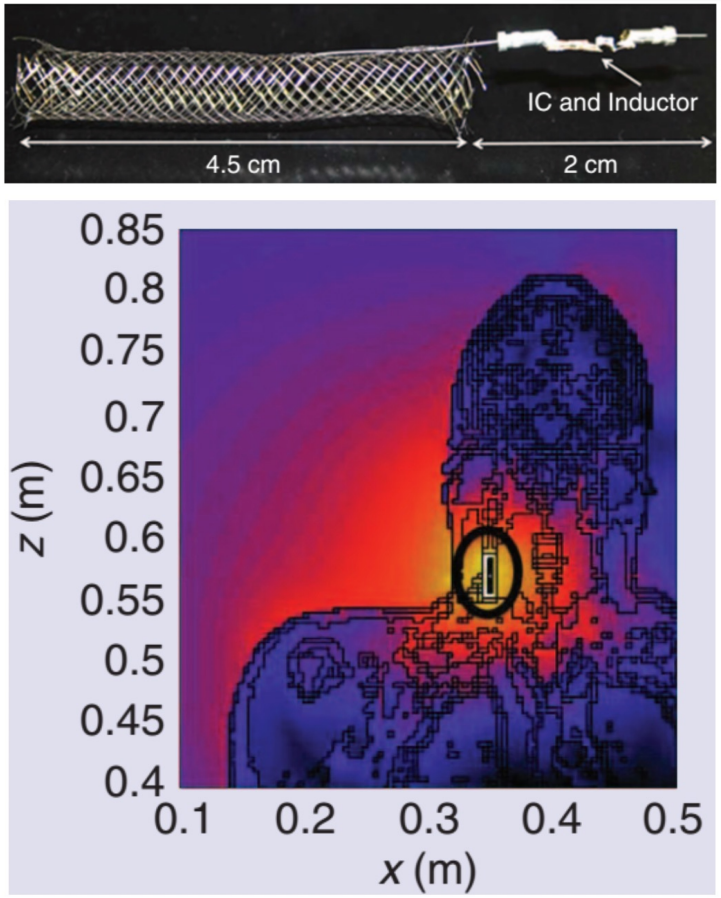
- The RFID main components: a programmable RFID tag for storing data, a reader with an antenna to read the tags, and an application software hosted on a computer to analyze the data.



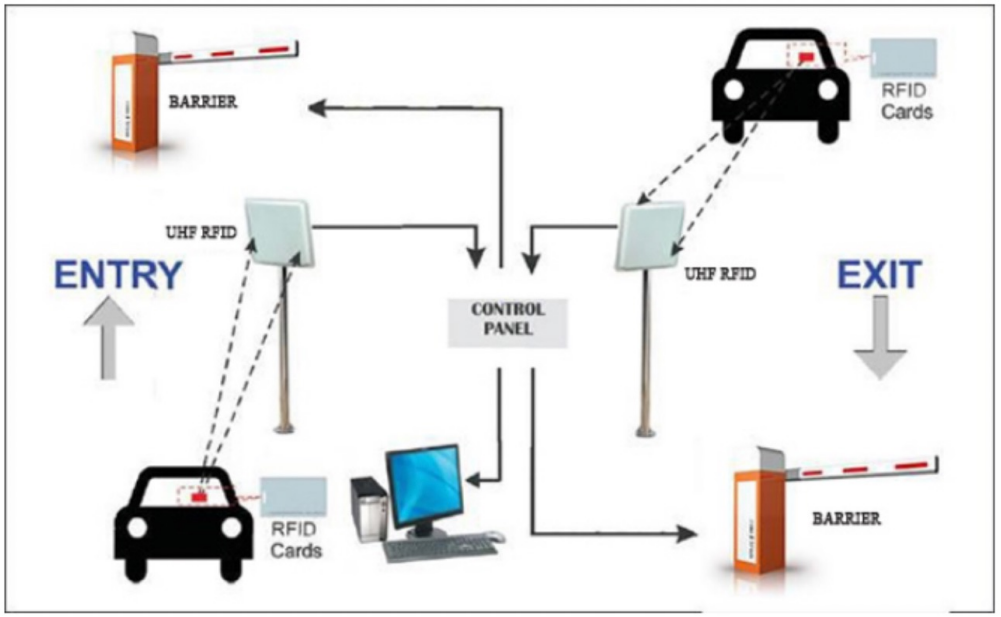


RFID main components

3.9. RFID



Design of implanted RFID tags for passive sensing of human body: The STENTag



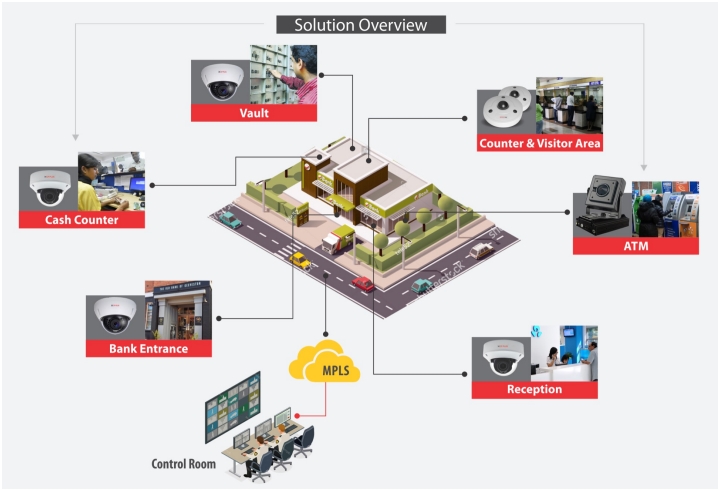
3.10. Video Tracking

- Video tracking is the process of capturing and analyzing the video feeds, frame by frame, of a particular object or person over a short time interval. It is used to measure and analyze movements, visual attention, as well as behavior. Video tracking is used for customer identification, surveillance, augmented reality, traffic control, and medical imaging.
- It is yet another mechanism to identify and monitor "things" when sensors or RFID tags are not available. Video tracking may also be used in conjunction with sensors and/or RFID to provide a more comprehensive solution.

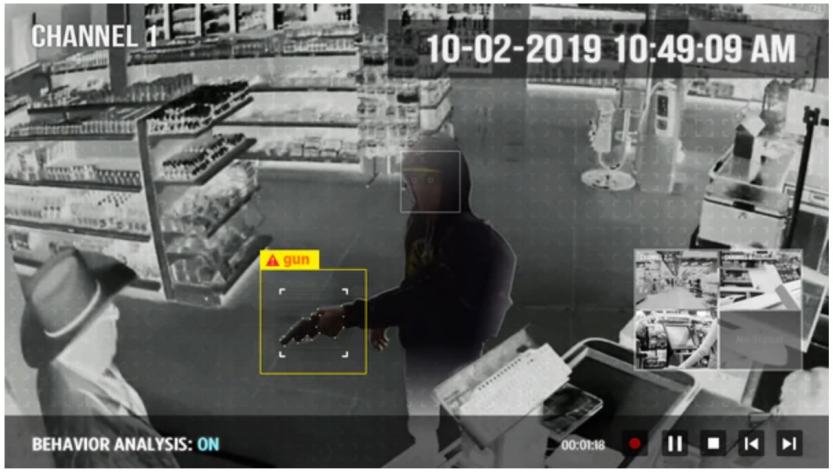
3.10. Video Tracking



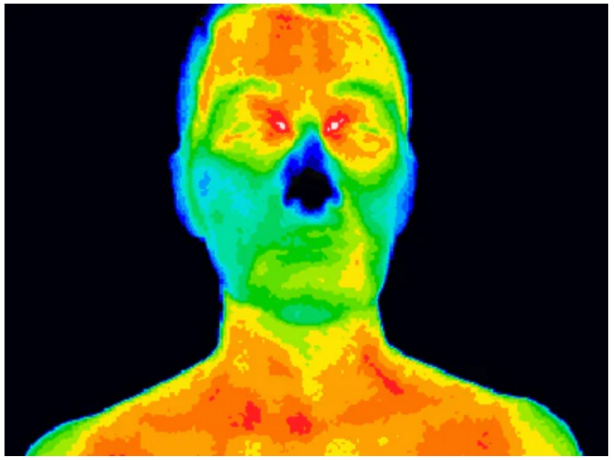
Video Analytics Applications for IoT Surveillance



Video Analytics Applications for Banking



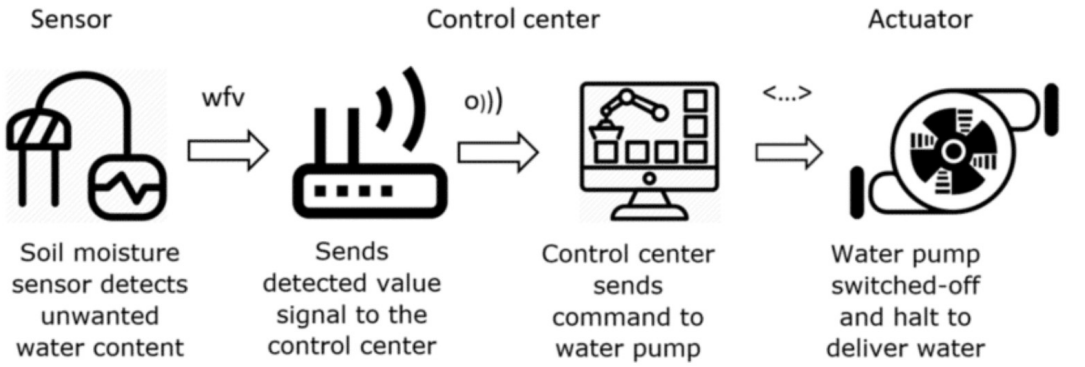
Behavioral Prediction



Medical Intervention

3.11. IoT Actuators

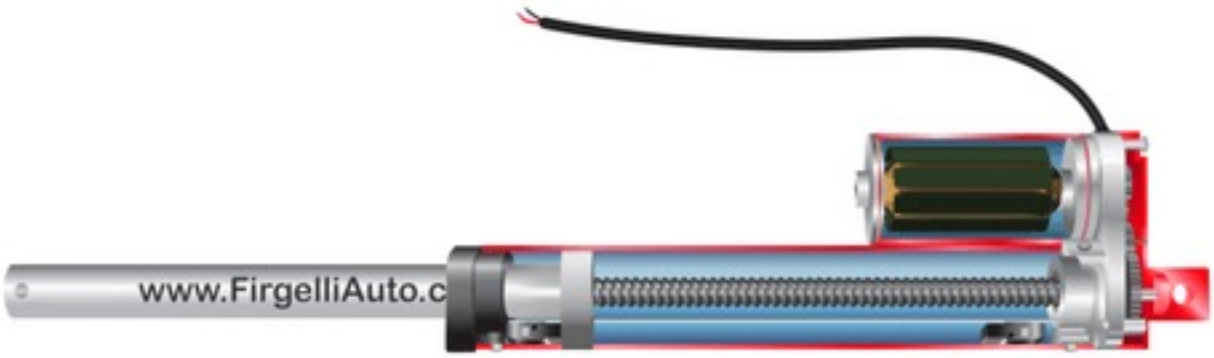
- An actuator is a type of motor that is responsible for controlling or taking action in a system. It takes a source of data or energy (e.g., hydraulic fluid pressure, other sources of power) and converts the data/energy to motion to control a system.





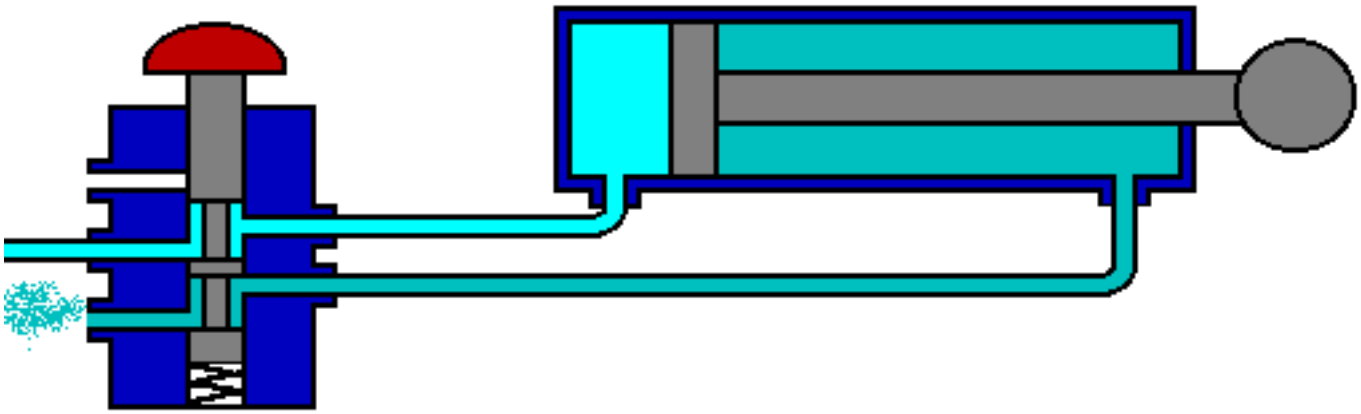
3.11. IoT Actuators

- ***Mechanical Linear Actuators***: Mechanical actuators convert rotary motion to linear motion. Devices such as screws and chains are utilized in this conversion. The simplest example of mechanical liner actuators is referred to as the “screw” where leadscrew, screw jack, ball screw, and roller screw actuators all operate on the same principle by rotating the actuator’s nut, the screw shaft moves in a line.



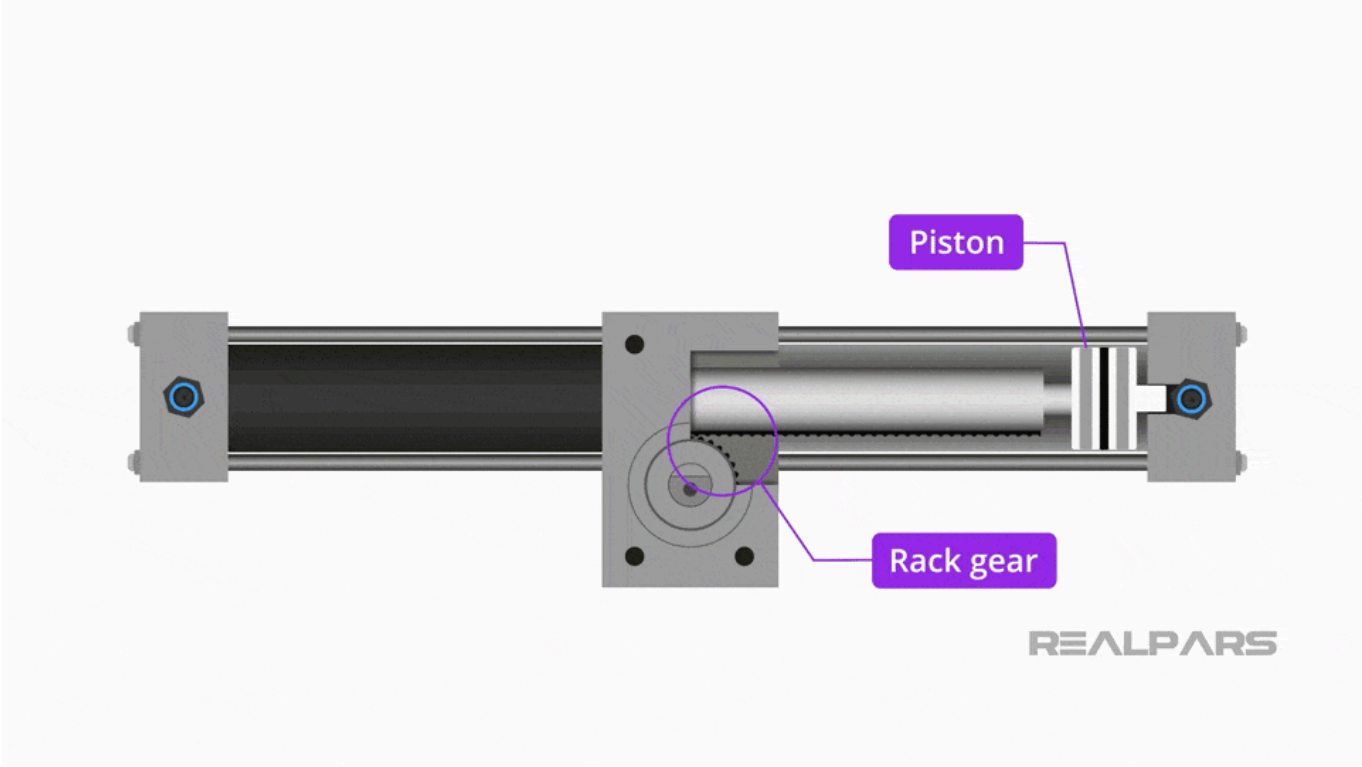
3.11. IoT Actuators

- ***Hydraulic Actuators***: Hydraulic actuators are simple devices with mechanical parts that are used on linear or quarter-turn valves. They are designed based on Pascal's Law: when there is an increase in pressure at any point in a confined incompressible fluid, then there is an equal increase at every point in the container.



3.11. IoT Actuators

- ***Pneumatic Actuators***: Pneumatic actuators work on the same concept as hydraulic actuators except compressed gas is used instead of liquid.



Thanks for your listening!

